

Single-cell transcriptomics reveals cell-type-specific circadian rhythms and their disruption by acute misalignment in mouse aorta

Benjamin J. Auerbach^{1,*}, Ronan Lordan^{2,*}, Soon Y. Tang², Seán T. Anderson², Elizabeth Hennessy², Ryan McConnell², Mingyao Li³, Garret A. FitzGerald²

^{*}These authors contributed equally to this work.

¹Graduate Group in Genomics and Computational Biology, University of Pennsylvania
Perelman School of Medicine, Philadelphia, PA 19104, USA

²Institute for Translational Medicine and Therapeutics, University of Pennsylvania
Perelman School of Medicine, Philadelphia, PA 19104, USA

³Department of Biostatistics, Epidemiology and Informatics, University of Pennsylvania
Perelman School of Medicine, Philadelphia, PA 19104, USA.

Correspondence:

Garret A. FitzGerald

Institute for Translational Medicine and Therapeutics

University of Pennsylvania Perelman School of Medicine

Room 10-122, Smilow Center for Translational Research, 3400 Civic Center Boulevard

Philadelphia, PA 19104

Email: garret@upenn.edu

Tel: (215) 898-1185

Fax: (215) 573-9135

Abstract

The circadian molecular clock is a 24-hour cellular timekeeper that influences many features of cardiovascular function. Disruption of the circadian clock via misalignment with the light-dark cycle is associated with a higher incidence of cardiovascular disease and raises cardiovascular risk factors in humans. Nonetheless, the cell-type-specific molecular basis for how misalignment affects the vasculature remains poorly understood. To address this, we performed single-cell RNA-sequencing from whole mouse aorta at ZT0, ZT6, ZT12, and ZT18 under aligned and acutely misaligned (6-hour phase advance) light-dark cycles in both male and female mice. Leveraging Bayesian variational inference, we estimated posterior waveforms for 141,752 cells across four major cell types and identified hundreds of confidently cycling genes in vascular smooth muscle cells (SMCs) and fibroblasts. Pathway and transcription factor enrichment analyses revealed coordinated circadian activity in cholesterol biosynthesis, muscle contraction, and extracellular matrix organization. Notably, SMC genes implicated in phenotypic switching showed coordinated temporal patterns, with genes promoting switching peaking at dusk and genes restraining switching peaking at dawn. Comparing males and females, we found that female SMCs are broadly more rhythmic, with higher amplitudes and nearly twice as many cycling genes after controlling for cell counts and library sizes—a sex difference that was cell-type-specific and not observed in fibroblasts. After acute misalignment, cycling genes showed reduced amplitudes, and peak times showed limited adaptation to the new light-dark cycle. Given that the central clock is known to adapt near-completely during this timeframe, these observations suggest internal misalignment between central and peripheral rhythms. Moreover, altered relative timing of core clock genes within cells indicates that misalignment is created at the intracellular level as well. In SMCs, gene expression patterns were consistent with proteostatic stress, including broad downregulation of protein chaperones and stress-response genes, which cells appear to cope with by upregulating protein degradation pathways. This atlas provides a resource for un-

derstanding how circadian misalignment disrupts vascular homeostasis and may contribute to cardiovascular disease risk.

Introduction

The circadian molecular clock is a 24-hour timekeeping mechanism found in nearly every cell in humans. The time of the clock, referred to as circadian phase, is determined by the mRNA and protein concentrations of the clock’s constituent genes, referred to as clock or core clock genes. Clock genes are organized in a transcriptional-translational feedback loop that enables cells to maintain self-sustained $\sim$ 24-hour oscillations in the concentrations of clock gene mRNA. Clock gene proteins additionally interact with cell-type-specific regulatory factors to drive rhythmic transcription of genes referred to as clock-controlled genes (CCGs). It is largely through these CCGs that circadian clocks generate rhythmic cellular and organismal behaviors.

Many features of cardiovascular function are under direct control of the circadian clock, such as vascular tone [1], thrombus formation [2], endothelial nitric oxide synthase activity [3], and myeloid cell recruitment and adhesion [4, 5]. Crucially, disruption of the circadian clock modulates vascular disease. Disruption of the clock via light-dark (LD) misalignment by forced cycle desynchrony protocols increases cardiovascular risk factors in humans [6]. Epidemiological studies associate shift work with an increased incidence and prevalence of cardiovascular disease [7, 8, 9]. Genetic ablation of the clock in mice influences atherosclerosis via multiple orthogonal mechanisms, such as effects on vascular smooth muscle cell (SMC) migration [10], SMC phenotypic switching and plaque stability [11], lipid metabolism [12], and rhythmic myeloid recruitment to the vasculature [4]. Moreover, existing evidence suggests clock function of the endogenous vascular cell types restrains atherosclerosis [13]. Nonetheless, the cell-type-specific molecular basis by which circadian misalignment affects the vasculature remains incompletely understood.

In humans, biomarkers of cardiovascular risk differ between males and females. These include arterial stiffness [14], lipid profiles [15, 16], blood pressure [17], and immune phenotypes [18]. The presentation of vascular disease also differs between sexes, such as differences in atherosclerotic plaque size, composition, and stability [19, 20]. Existing literature suggests females exhibit more robust circadian characteristics than males and their clocks appear more resilient to disruption [21]. Considering this, and given the clock’s influence over key vascular disease risk factors [22, 23, 24], characterizing how the sexes differ in their molecular response to clock perturbations will be important to understand whether sex differences in vascular disease presentation are mediated by the circadian clock. Here, we use single-cell RNA-sequencing to generate a cell-type-specific circadian atlas of the mouse aorta, characterize sex-dependent differences in circadian rhythmicity, and define the cell-type-specific transcriptomic consequences of acute circadian misalignment.

Results

Generation of a single-cell mouse aorta circadian atlas

To investigate these questions, we performed several single-cell RNA-sequencing (scRNA-seq) experiments from whole aorta samples harvested over circadian time courses (Figure 1a). First, to assess the contributions of sex and cell type on circadian expression of cells aligned with the light-dark (LD) cycle, whole aorta samples were collected from 12-week-old male and female C57BL/6 mice entrained to a 12:12 LD cycle for 2 weeks. Mice were sacrificed at Zeitgeber time (ZT) 0, 6, 12, or 18 hours, and whole aorta samples were harvested (Figure 1b). We additionally collected data from inducible *Bmal1* global deletion models (Methods); these results are presented in the Supplementary Results. To assess the response to circadian misalignment, 12-week-old male and female C57BL/6 mice were entrained to a 12:12 LD cycle for 2 weeks, after which the start of the LD cycle was advanced 6 hours (Figure 1c). Mice were sacrificed 5 days after the phase advance at ZT 0, 6, 12, or 18 hours.

For all conditions, cells were dissociated into single-cell suspensions and profiled using the 10X Genomics Chromium platform.

In total, we generated 152,473 cells. After quality control for library size (minimum 1000 UMIs), mitochondrial UMI fraction (maximum 0.2), and doublet removal using Scrublet [25], 141,752 high-quality cells remained. To obtain a low-dimensional embedding, 1150 highly variable genes were detected (Methods) and used as input to scVI [26], with library size, Bmal1 knockout status, misalignment status, batch, sex, and time as covariates. Cells were clustered using the Leiden algorithm [27] with resolution 0.05, yielding 5 main clusters: 115,460 vascular smooth muscle cells (SMCs), 19,759 fibroblasts, 3,935 endothelial cells (ECs), 2,917 macrophages, and 631 T cells (Figures 1d, 1f). SMCs were further subclustered into 5 subtypes (Figures 1e, 1g). The majority of SMCs expressed canonical contractile markers, with variation primarily reflecting physical location: SMC0 (Hoxb5+, Hoxb6+) and SMC1 (Lgals3+, Hand2+) expressed markers of the aortic arch and ascending aorta [28, 29, 30], while SMC2 (Hoxa10+, Hoxc10+, Hoxa9+, Rgs5+) expressed markers of the descending thoracic aorta [28, 29, 30, 31]. Smaller subpopulations expressed markers of synthetic phenotypes: SMC3 expressed osteoblast [32, 33] (Runx2, Frzb) and chondrocyte [34] (Sox9) markers, while SMC4 expressed markers of multipotent SEM cells [35] (Ly6a, Ly6c1, Vcam1).

To estimate circadian expression parameters, approximate posterior distributions of gene temporal waveforms were inferred using Bayesian variational inference for each gene, cell type, and condition combination (Methods). Circadian cycling genes were identified by decomposing posterior waveforms using a Fast Fourier Transform (FFT) and computing the expected decrease in data likelihood when assuming the 24-hour component amplitude was zero (Methods). As a data quality check, we assessed whether the peak times of SMC cycling genes in our single-cell data aligned with peak times from bulk aorta transcriptomics, given that SMCs comprise most of the vessel wall. The posterior acrophases of SMC cycling

genes aligned well with their peak times from bulk aorta generated by Zhang et al. [36] (Supplementary Figure 1a). As anticipated, core circadian clock genes demonstrated strong rhythms across all major cell types (Supplementary Figure 1b) and SMC subtypes (Supplementary Figure 1c). The acrophase distribution of cycling genes appeared bimodal, with peaks at dawn and dusk (Supplementary Figure 1d). Data from Bmal1 deletion mice revealed virtually no cycling genes beyond those driven by the core clock, suggesting the vast majority of detected cycling in wild-type mice reflects clock-controlled transcription. Altogether, these analyses provided confidence in both our data quality and our approach to calling cycling genes and estimating their parameters.

Cell-type-specific daily cycling expression in the mouse aorta

To identify genes with robust daily cycling shared between sexes, we intersected male and female cycling gene lists within each cell type, requiring acrophases to be within 3 hours. This yielded 416 shared daily cyclers in SMCs, 117 in fibroblasts, 21 in ECs, and 5 in macrophages (Figure 2a).

To identify pathways enriched among daily cyclers, we performed Reactome pathway enrichment analysis, restricting to gene sets whose members peak at similar times (Methods). In SMCs, the most strongly enriched pathways included cholesterol biosynthesis (peaking at dusk), metabolism of steroids, metabolism of carbohydrates, Wnt signaling, muscle contraction [41, 42, 43], PPAR α -regulated lipid metabolism, circadian clock, apoptosis, and degradation of the extracellular matrix (Figure 2b). As has been previously described [41], many features of aortic metabolism undergo circadian rhythms. In fibroblasts, daily cyclers were enriched for extracellular matrix organization (peaking at dawn), circadian clock, and PPAR α -regulated lipid metabolism (Figure 2b). EC and macrophage cyler counts were too low for robust pathway enrichment. We additionally performed transcription factor (TF) motif enrichment, identifying gene sets whose members share promoter motifs and peak at coordinated times. In SMCs, enriched motifs included FOXO3A, Nkx3-2, SRF, and AP1

(Figure 2c). In fibroblasts, enriched motifs included NR1H3, SP3, MYC, EGR1, SP1, and TEAD4, with most target genes peaking near ZT12 (Figure 2c). Male-specific and female-specific daily cycling analyses, including pathway enrichment and SMC phenotypic switching, are shown in Supplementary Figures 2 and 3.

We next assessed whether daily cycling expression was relevant to SMC phenotypic switching in atherosclerosis. SMCs can dedifferentiate into stem cell and endothelial-like mesenchymal (SEM) cells, which are multipotent and can differentiate into macrophage-like and fibroblast-like cells [35, 37, 38, 39]. Daily cyclers in SMCs that monotonically increase from SMC to SEM tended to peak at dusk (ZT12), while those that decrease from SMC to SEM tended to peak at dawn (ZT0) (Figure 2d). Though circadian gating of cell fate decisions has been observed in numerous contexts [69, 70, 71], it remains unknown whether the clock gates SMC phenotypic switching.

Enhanced rhythmicity of female aortic cell types

The number of confident circadian cycling genes was roughly comparable between male and female cell types, with modestly more cyclers detected in females. Female circadian gene expression was primarily distinguished by higher amplitude rhythms (Figure 3a). Moreover, among cycling genes common to both sexes, acrophases tended to peak modestly earlier in females, consistent with previous observations [21]. To control for cell count and library size discrepancies, we downsampled male and female cells and pseudobulk UMI counts to match within each cell type before rerunning our Bayesian regression pipeline (Methods). After this adjustment, female SMCs retained nearly twice as many cycling genes as males across Bayes factor thresholds, while fibroblasts showed comparable numbers between sexes (Figure 3b). These results suggest sex-dependent rhythmicity is cell-type-specific. Macrophages additionally demonstrated stronger female rhythmicity, though this result should be interpreted cautiously given the low cell count.

To identify pathways and transcription factor networks with sex-differential temporal expression, we performed a two-group waveform enrichment analysis, computing the difference waveform (female – male) for each gene set and testing whether it deviates from zero at each time point (Methods). In SMCs, gene sets involved in antigen processing and presentation, spliceosome, protein processing in the endoplasmic reticulum, endocytosis, and estrogen signaling showed lower expression in females, particularly at ZT6 and ZT12 (Figure 3c). Among transcription factor networks, targets of SMAD3, Nkx2-5, and SMAD4 showed sex-differential temporal patterns, with lower female expression concentrated at ZT6 (Figure 3c). In fibroblasts, NF- κ B signaling, ECM-receptor interaction, and TNF signaling gene sets showed the largest amplitude sex differences (Figure 3c). Extended Reactome and GO Biological Process waveform enrichment results are shown in Supplementary Figure 4.

Circadian misalignment disrupts proteostasis and cell-type entrainment in the aorta

After a 6-hour phase advance, the number of confident cycling genes decreased across all cell types, even after controlling for cell count and library size differences (Figure 4a; Supplementary Figure 5). Among cycling genes common to both conditions, acrophases were generally delayed in misaligned mice, indicating incomplete entrainment to the new LD cycle (Figure 4b). Amplitudes of shared cyclers were also generally decreased. The central clock in the suprachiasmatic nucleus (SCN) is known to re-entrain to a 6-hour phase advance within approximately 5 days, meaning the master clock had likely adapted to the new LD cycle by the time of tissue collection. The incomplete re-entrainment of vascular cell types at this time point implies internal desynchrony between the central clock and the peripheral vasculature — a state that may extend to other peripheral organs and has been proposed to underlie many of the adverse health consequences of jet lag and shift work.

Beyond desynchrony between the central and peripheral clocks, misalignment also appeared

to alter the intracellular timing relationships among clock genes within SMCs. Restricting to clock genes whose posterior acrophase 95% interval was within 2 hours in both conditions, we found that *Dbp* and *Nr1d1* — both direct transcriptional targets of BMAL1:CLOCK — compressed their phase delay relative to *Arntl* by approximately 1.3 and 1.2 hours, respectively (Figure 4c). This suggests that even within individual cell types, the internal architecture of the clock may be rearranged during acute misalignment. Acute misalignment also appeared to alter macrophage recruitment timing. While most shared cycling genes shifted acrophases toward the new LD cycle, macrophage *Ccr2* and *Cx3cr1* — which mediate rhythmic macrophage recruitment to the vasculature [4, 89, 90, 91] — showed acrophase delays of approximately 6 hours, suggesting minimal adaptation (Figure 4d). However, macrophage estimates should be interpreted cautiously given the low cell counts and shallow sequencing depth in this population.

The most striking effect of misalignment in SMCs was disruption of proteostasis gene expression (Figure 4e,f). Multiple chaperone genes — including *Hspa1a*, *Hspa1b*, *Dnajb1*, *Dnajb4*, *Dnaja1*, and *Hsph1* — were substantially downregulated, particularly at ZT0 (Figure 4f). Beyond their roles in heat shock response, these chaperones are implicated in endoplasmic reticulum (ER) stress and the unfolded protein response. Consistent with disrupted proteostasis, *Xbp1*, a key regulator of the unfolded protein response implicated in preventing neointimal hyperplasia [105], was downregulated at ZT0, while *Ankrd1*, which protects cells from ER stress [106], was downregulated during the light phase (Figure 4f). *Paip2b*, which inhibits mRNA translation [107], was downregulated at all time points, potentially contributing to the accumulation of unfolded proteins. Conversely, *Uba52*, which targets proteins for degradation [108, 109], was strongly upregulated (Figure 4f), and gene sets involved in transcription and ribosome biogenesis showed decreased mesors — possibly reflecting compensatory efforts to reduce the unfolded protein burden. *Atf3*, which has been described to inhibit SMC phenotypic switching, was substantially decreased at ZT0 with an acrophase delay of approximately 7 hours (Figure 4f), suggesting acute misalignment may

additionally predispose SMCs to phenotypic switching. Female-specific misalignment effects are presented in the Supplementary Results.

Discussion

Circadian clock disruption has a notable influence on cardiovascular health, yet the cell-type-specific molecular basis by which misalignment with the light-dark cycle affects the vasculature remains poorly understood. Bulk RNA-sequencing circadian time courses of the aorta have defined key vascular functions that cycle with circadian rhythms [41], but cannot resolve cell-type-specific contributions. Here, we used scRNA-seq to generate a cell-type-specific circadian atlas of the mouse aorta, characterize sex differences in rhythmicity, and define the transcriptomic consequences of acute circadian misalignment.

Our atlas reveals that the extent of circadian transcription varies markedly across vascular cell types. SMCs and fibroblasts harbor hundreds of confidently cycling genes, while ECs and macrophages show far fewer — likely reflecting both biological differences in clock output and lower statistical power from smaller cell populations. Pathway enrichment among SMC cyclers highlighted coordinated rhythms in cholesterol biosynthesis, muscle contraction, extracellular matrix organization, and lipid metabolism, consistent with prior bulk aorta studies [41, 42, 43]. An intriguing finding was the temporal coordination of genes implicated in SMC phenotypic switching: genes that increase during the SMC-to-SEM transition peaked at dusk, while those that decrease peaked at dawn. Although circadian gating of cell fate decisions has been observed in other contexts [69, 70, 71], whether the clock gates SMC phenotypic switching remains an open question warranting functional investigation.

Comparing males and females, we found that sex differences in rhythmicity were cell-type-specific rather than uniform. Female SMCs exhibited higher amplitude rhythms and nearly twice as many cycling genes after controlling for cell counts and library sizes, whereas fibroblasts showed comparable rhythmicity between sexes. These findings are consistent with

previous observations that females exhibit more robust circadian characteristics [21] and extend them by demonstrating that this effect is not a general property of all vascular cell types. The molecular basis of this cell-type specificity is unclear, but future work integrating sex hormone perturbations with circadian profiling may help disentangle the contributions of hormonal signaling and cell-intrinsic clock properties.

The most notable findings of our study concern the effects of acute circadian misalignment, which appeared to disrupt circadian organization at multiple levels. At the organ level, vascular cell types showed incomplete re-entrainment to the new LD cycle 5 days after the phase advance. Given that the central clock in the SCN is known to re-entrain within this timeframe, this implies internal desynchrony between central and peripheral clocks — a state proposed to underlie many adverse health consequences of jet lag and shift work [6]. At the intracellular level, the phase relationships among clock genes were altered: *Dbp* and *Nr1d1* compressed their phase delay relative to *Arntl*, suggesting the internal architecture of the molecular clock is rearranged during acute misalignment, not merely shifted. At the intercellular level, tissue-resident macrophage expression of *Ccr2* and *Cx3cr1* — which participate in signaling axes that mediate rhythmic myeloid recruitment to the vasculature [4, 89, 90, 91] — showed minimal adaptation to the new LD cycle, in contrast to the partial re-entrainment observed in endogenous vascular cell types. This differential adaptation may disrupt the coordinated timing between macrophage and vascular cell type signaling, though the functional consequences of such intercellular desynchrony remain to be tested.

Beyond these timing disruptions, the most striking molecular consequence of misalignment in SMCs was disruption of proteostasis. Multiple chaperone and stress-response genes were substantially downregulated, including key components of the HSP70 system (*Hspa1a*, *Hspa1b*, *Dnajb1*, *Dnajb4*, *Dnaja1*, *Hsph1*) and regulators of the unfolded protein response (*Xbp1*, *Ankrd1*). Downregulation of the translational repressor *Paip2b* may exacerbate this burden by permitting continued protein synthesis. Conversely, upregulation of *Uba52*, which

targets proteins for proteasomal degradation [108, 109], and decreased mesors of gene sets involved in transcription and ribosome biogenesis may reflect compensatory efforts to reduce the accumulation of misfolded proteins. Disrupted proteostasis has been implicated in vascular pathology, including ER stress-driven neointimal hyperplasia [105] and atherosclerotic plaque instability. Whether acute misalignment indeed increases the unfolded protein burden, and whether these compensatory responses are sufficient to maintain proteostasis, will require functional follow-up. Additionally, the substantial downregulation of Atf3, which has been described to inhibit SMC phenotypic switching, suggests that acute misalignment may predispose SMCs to dedifferentiation — a possibility reinforced by the coordinated temporal patterns of switching genes observed in aligned mice.

Our study has several limitations. While we collected many SMCs, allowing high-confidence circadian parameter estimation, we captured relatively few ECs and macrophages. The number of confident cyclers in these populations is likely an underestimate, and macrophage-specific findings — including the intercellular timing observations — should be interpreted cautiously given the low cell counts and shallow sequencing depth. Additionally, our scRNA-seq data capture transcript-level changes only; the proteostasis disruption we infer from gene expression patterns requires biochemical validation. Our study also captures a single time point after misalignment (5 days post-advance), and the trajectory of adaptation — whether these effects resolve, persist, or worsen — remains unknown.

Altogether, this atlas provides a resource for understanding how the circadian clock organizes vascular gene expression across cell types and sexes, and how acute misalignment disrupts this organization at multiple scales — from intracellular clock architecture to intercellular coordination to proteostatic homeostasis. These findings may help explain why circadian misalignment elevates cardiovascular risk and highlight proteostasis as a candidate mechanism linking circadian disruption to vascular disease.

Figures

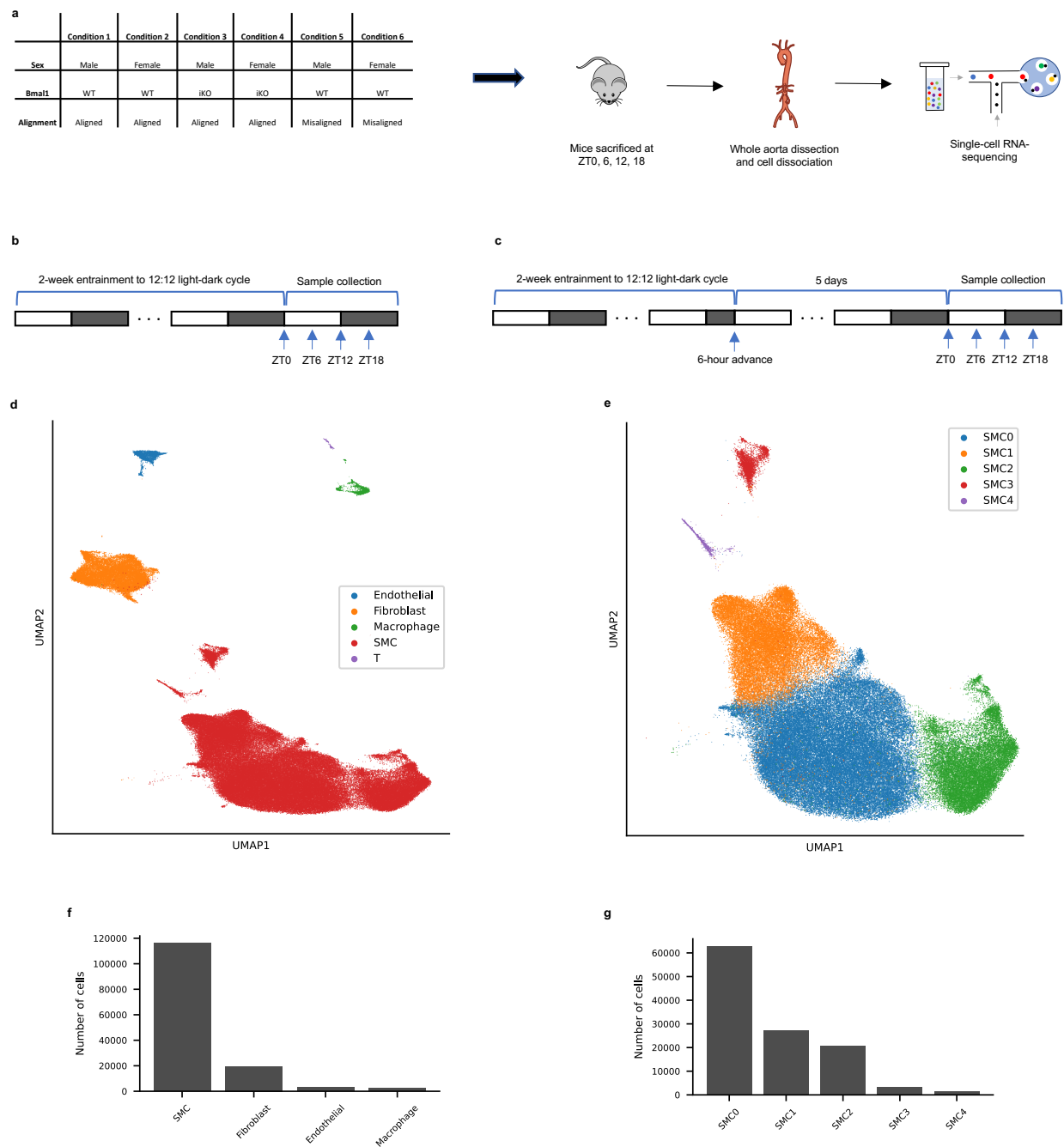


Figure 1

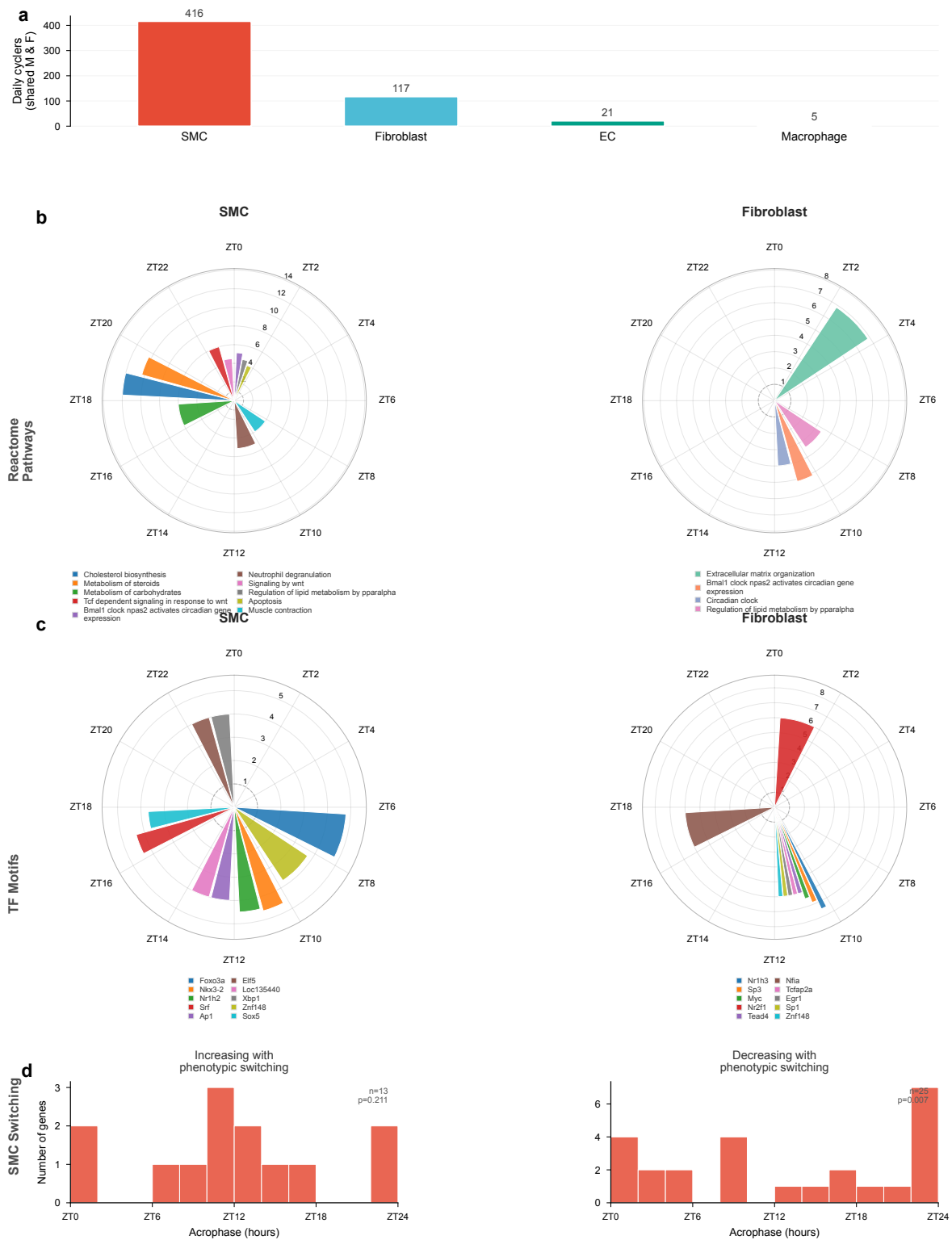


Figure 2

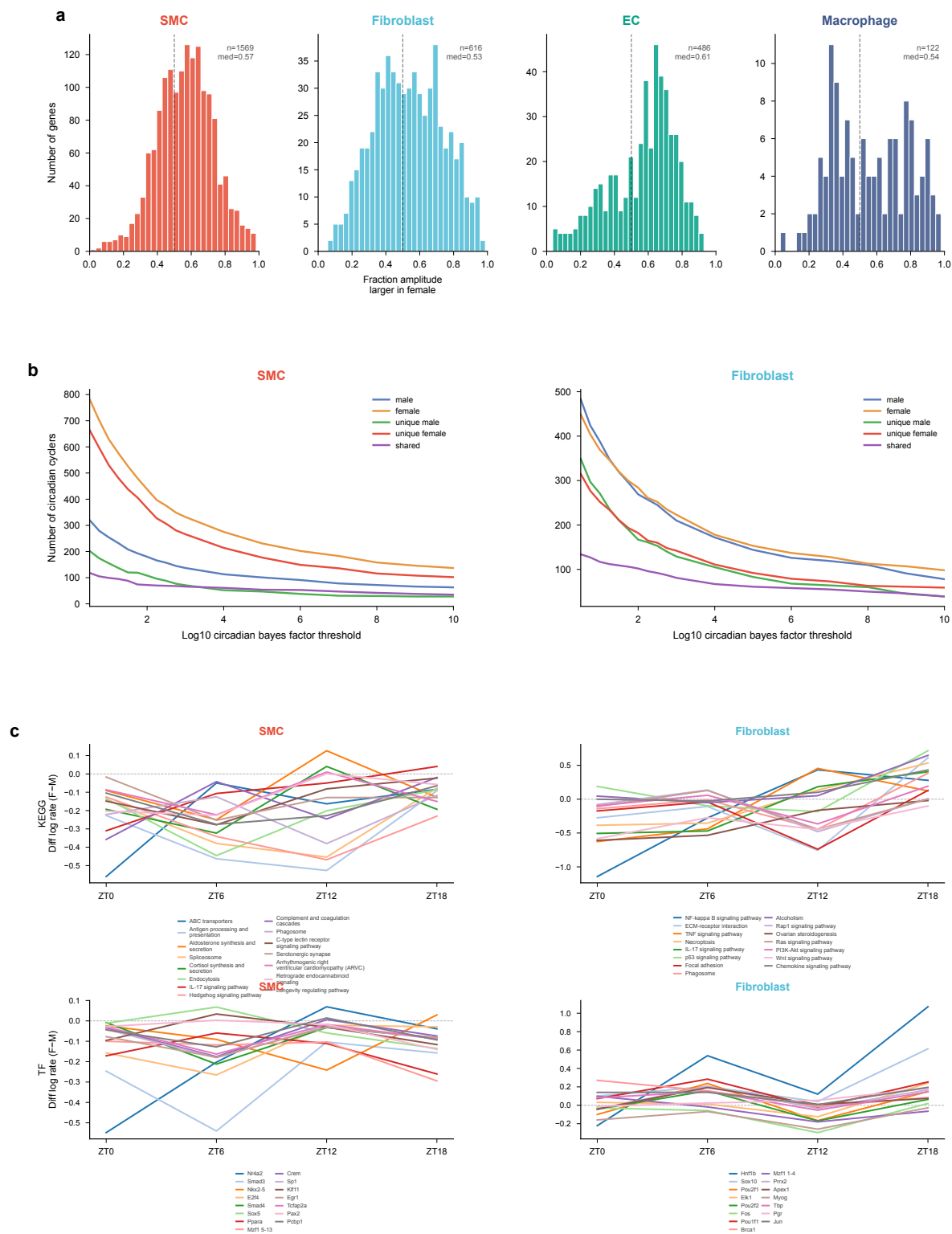


Figure 3

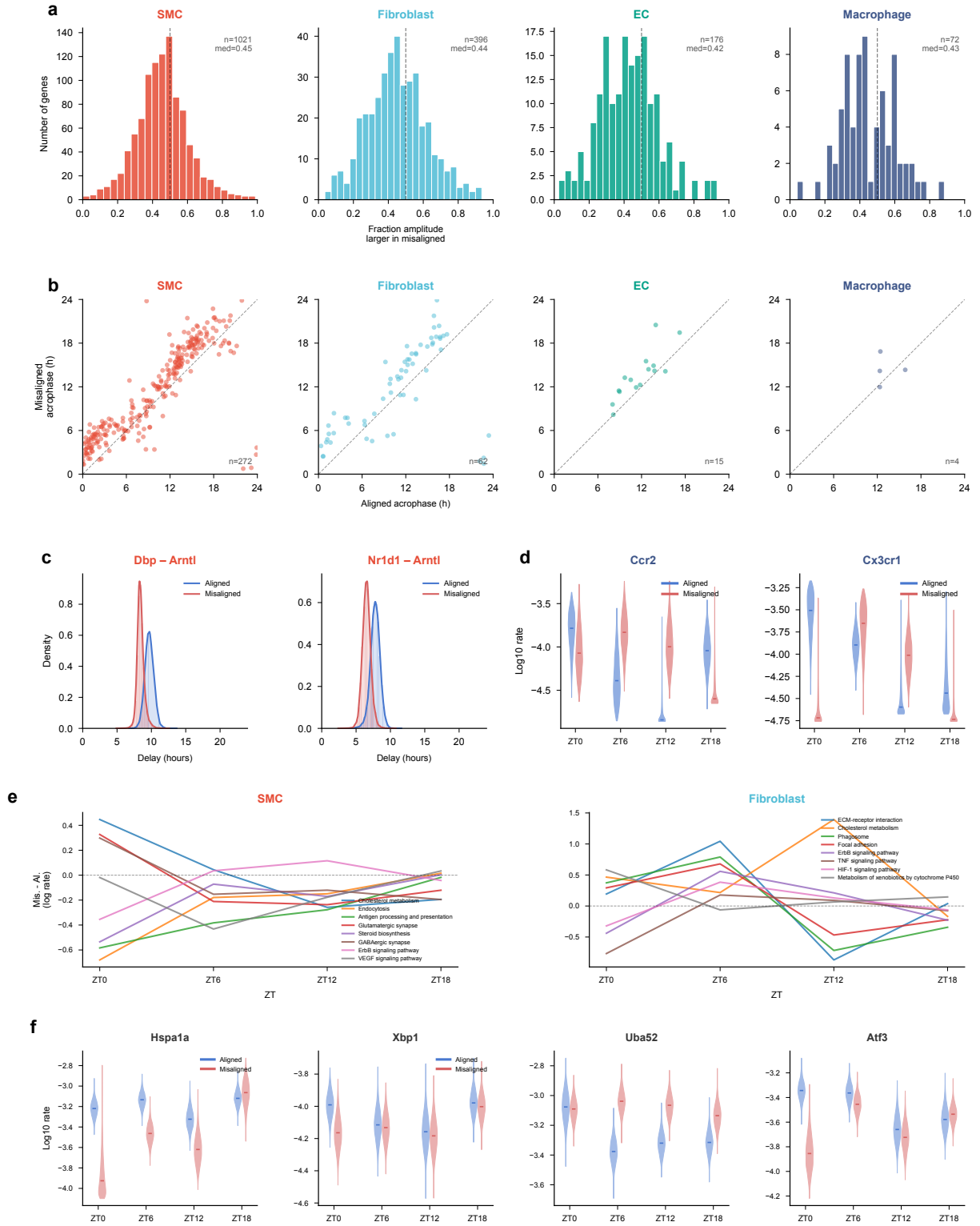


Figure 4

Figure Legends

Figure 1: Experimental overview and cell types of the mouse aorta circadian atlas. a) Experimental conditions: male and female C57BL/6 mice under aligned and misaligned light-dark cycles, and male and female $Bmal1^{fl/fl};CAGG^{creERt2/+}$ mice under aligned conditions. Mice were sacrificed at ZT0, ZT6, ZT12, and ZT18, whole aortas were dissected and dissociated, and single-cell RNA-sequencing was performed. b) Light-dark cycle and sample collection protocol for aligned mice. c) Light-dark cycle and sample collection protocol for misaligned mice, subjected to a 6-hour phase advance 5 days prior to sacrifice. d) UMAP visualization of scVI latent embedding for all cells, colored by cell type. e) UMAP visualization of SMCs, colored by subtype. f) Number of cells per major cell type cluster. g) Number of cells per SMC subtype cluster.

Figure 2: Cell-type-specific daily cycling expression in the mouse aorta. a) Number of daily cycling genes shared between males and females in each major cell type, requiring acrophases within 3 hours. b) Reactome pathway enrichment among daily cyclers in SMCs (left) and fibroblasts (right). Rose plots show enriched gene sets positioned by the mean acrophase of their member genes; radial extent indicates the number of member genes. c) Transcription factor motif enrichment among daily cyclers in SMCs (left) and fibroblasts (right), plotted as in (b). d) Acrophase distributions of SMC daily cyclers implicated in phenotypic switching. Left: genes that monotonically increase during the SMC-to-SEM transition. Right: genes that monotonically decrease. n and p -values from permutation enrichment tests for dusk (left) or dawn (right) peak times.

Figure 3: Enhanced rhythmicity of female aortic cell types. a) Distribution of the fraction of posterior amplitude samples larger in females than males, for genes cycling in either sex. Dashed line indicates 0.5 (no difference); median shown. Distributions shifted right indicate higher female amplitudes. b) Number of circadian cycling genes in males and females across

$\log_{10}$ Bayes factor thresholds, after downsampling cells and pseudobulk UMI counts to match between sexes. SMCs (left) and fibroblasts (right). c) Sex-differential waveform enrichment analysis. Difference waveforms (female – male) for KEGG pathway gene sets (top) and transcription factor motif gene sets (bottom) in SMCs (left) and fibroblasts (right). Each line represents a gene set; values above zero indicate higher expression in females at that time point.

Figure 4: Circadian misalignment disrupts proteostasis and cell-type entrainment in the aorta. a) Distribution of the fraction of posterior amplitude samples larger in misaligned than aligned mice, for genes cycling in either condition. Distributions shifted left of 0.5 indicate reduced amplitudes after misalignment. b) Acrophase scatter plots comparing aligned (x-axis) and misaligned (y-axis) peak times for cycling genes shared between conditions. Dashed line indicates identity (perfect entrainment); points above the line indicate delayed acrophases. n , number of shared cycling genes. c) Posterior distributions of the phase delay between *Dbp* and *Arntl* (left) and *Nr1d1* and *Arntl* (right) in SMCs, comparing aligned and misaligned conditions. Compression of the delay indicates altered intracellular clock architecture. d) Posterior waveforms of macrophage *Ccr2* (left) and *Cx3cr1* (right) in aligned and misaligned conditions. Violin plots show posterior $\log_{10}$ transcript proportion estimates at each time point; horizontal lines indicate posterior means. e) KEGG pathway waveform enrichment showing the difference in $\log_{10}$ expression rate (misaligned – aligned) at each time point for SMCs (left) and fibroblasts (right). Values below zero indicate decreased expression after misalignment. f) Posterior waveforms of representative proteostasis-related genes in SMCs, comparing aligned and misaligned conditions. Violin plots as in (d).

Methods

Inducible deletion of the Bmal1 gene in Bmal1^{fl/fl}:CAGG^{creERT2/+} mice

At 8 weeks of age, Bmal1 was inducibly deleted in Bmal1^{fl/fl}:CAGG^{creERT2/+} mice by administering tamoxifen via oral gavage (100 mg/kg of body weight) for 5 days. Confirmation of Bmal1 protein expression loss was confirmed via Western blot ([Supplementary Figure 6](#)).

Aorta collection and scRNA-seq data generation

For the aligned conditions, 12-week-old C57BL/6J and Bmal1^{fl/fl}:CAGG^{creERT2/+} male and female mice from the Jackson Laboratory were entrained to a 12:12 LD cycle (12 hours of light, followed by 12 hours of darkness) for 2 weeks in circadian boxes with ventilation (20–22°C, RH ~50%) and fed ad libitum. At ZT0, ZT6, ZT12 (lights off), or ZT18 mice were sacrificed (2 per time point) via CO₂ exposure.

For the misaligned conditions, 12-week-old C57BL/6J male and female mice from the Jackson Laboratory were entrained to a 12:12 LD cycle (12 hours of light, followed by 12 hours of darkness) for 2 weeks in circadian boxes with ventilation (20–22°C, RH ~50%) and fed ad libitum. Mice were then phase advanced 6 hours. 5 days after the phase advance, mice were sacrificed at ZT0, ZT6, ZT12 (lights off), or ZT18 (2 per time point).

For each mouse in each condition, whole aorta was dissected and cut into small pieces (~1 mm), and cells were disassociated with 1.5 mL of enzyme cocktail (DNase–120 U/mL (Worthington, #LS006331), Liberase TM–4 U/mL (Roche, #05401127001), hyaluronidase–60 U/mL (Sigma-Aldrich, #H3506)) in a petri dish at 37°C for 40 mins. A pipette (1 mL) was used to dissociate aortic cells from the tissue pieces every 10 mins during the incubation. Cell supernatant was filtered through a 40 μ m strainer and washed with RPMI1640 (Gibco, #1187-085) containing 10% fetal bovine serum (FBS, HyClone, #SH30071.03) to inactive the enzyme cocktail. Residual red blood cells were lysed by incubating the cell suspension

with Red Blood Cell Lysing Buffer Hybri-Max (Sigma-Aldrich, #R7767) at RT for 1 min. The cells were washed two more times to remove debris with FACS buffer (FBS 2%), EDTA (5 mM, Invitrogen, #15575-038), HEPES (20 mM, Gibco, #15630-080), sodium pyruvate (1 mM, Gibco, #11360-070) in 1x PBS (Gibco, #14190-136), and resuspended in DMEM/F12 (Gibco, #11320-033) media containing 10% FBS for further analysis. Cells from mice sacrificed at the same time point were pooled to form single-cell suspensions. Single cells were then captured and barcoded using the 10X Genomics Chromium platform and sequenced using an Illumina NovaSeq S2 flow cell. Cell barcode detection, read alignment, and transcript quantification were performed using the 10X Genomics Cell Ranger pipeline.

Cell type marker gene detection procedure

Cell type markers used as input for low-dimensional embedding estimation were selected via a three-step procedure yielding 1969 marker genes.

First, markers whose variation defines major cell types were selected via a prior knowledge-based procedure. This was performed because preliminary clustering suggested high cell type imbalance of our dataset, with a disproportionate amount of SMCs and fibroblasts. As such, markers of rarer cell types (e.g. T cells) may not be identified using traditional methods based on detecting highly variable genes. Based on our prior work described in Pan et al. [35], we anticipate SMCs, fibroblasts, ECs, macrophages, and T cells. Based on this work, we selected several highly expressed and highly specific markers for each major cell type. We selected *Myh11* and *Acta2* for SMCs; *Serpinf1* and *Serping1* for fibroblasts; *Vcam1* and *Pecam1* for endothelial cells; *Cd68*, *Cd14*, *Cd16*, and *Cd64* for macrophages; and *Cd4*, *Cd8*, *Cd3g*, and *Cd3d* for T cells. For each major cell type, we then identified confident cells belonging to that cell type based on the transformed expression of their corresponding prior knowledge marker genes. Expression was transformed as follows:

$$X_{ij}^{(\text{transformed})} = \log_{10} \left(\min \left(\frac{X_{ij}}{L_i}, 10^{-7} \right) \right) \quad (1)$$

Where X_{ij} is the UMI count of gene j in cell i and L_i is the library size of cell i . Cells with transformed expression greater than -3 for any cell type marker was assigned to the corresponding cell type. This led to 119,817, 18,547, 3,632, 525, 185 cells roughly labeled as SMCs, fibroblasts, ECs, macrophages, and T cells. 5498 cells remained unlabeled.

Second, we identified markers with highly correlated expression to the prior knowledge cell type markers after the rough cell assignments. To identify these markers, the $\log_{10}$ pseudobulk relative expression of all genes was computed within each rough cell type. Cell type markers were then identified as those with $\log_{10}$ pseudobulk relative expression of at least -4 (i.e. 1 in 10,000 transcripts in pseudobulk) and with $\log_{10}$ pseudobulk relative expression larger than all other cell types by at least 0.5. This yielded 114, 113, 101, 228, 93 additional markers for SMCs, fibroblasts, ECs, macrophages, and T cells, respectively.

Third, we identified markers with high variability within cell types. To identify these marker genes, we first limited ourselves to genes that were sufficiently highly expressed. We decided cell type marker genes should be expressed at relative proportions of 1 in 10,000 transcripts in the cell types they mark. Moreover, there should be at least 30 cells observed for each cell type. Given this and given the median library size of 12,553 UMIs in our dataset, cell type marker genes should have at least 37 pseudobulk UMI for the cell type of interest (i.e. $10^{-4} \times 30 \times 12553$). Using this set of genes, we next restricted genes to those that had highly variable expression. Using the transformed expression counts, we computed the mean and variance of each genes' transformed expression. After fitting a nonparametric kernel (Python statsmodels implementation of Nadaraya-Watson, using bandwidth of 0.1 and default values for all other parameters) to describe the mean-variance relationship, genes with large variances were detected as those with Pearson residuals larger than 0.5.

Bayesian temporal regression model

Likelihood model

A Bayesian ANOVA regression model was fit to describe gene waveforms and detect genes with circadian rhythms. Our model was fit independently for each cluster and condition combination. As input, our model requires an $n \times p$ transcript count matrix $\mathbf{X}$, where n is the number of cells and p is the number of genes. For gene j in cell i with library size L_i and cell phase Θ_i (where $\theta_i \in \{1, \dots, q\}$ discrete time points), the unique molecular identifier (UMI) count X_{ij} was assumed to follow a Negative Binomial (NB) distribution:

$$X_{ij} \mid (\Theta_i = k) \sim \text{NB}(L_i \lambda_{jk}, \delta_{ij}) \quad (2)$$

Where k is the discrete index of the time the cell was collected at,

$$\mathbb{E}[X_{ij}] = L_i \lambda_{jk}, \quad (3)$$

$$\text{Var}(X_{ij}) = \mathbb{E}[X_{ij}] + \delta_{jk} \times (\mathbb{E}[X_{ij}])^2, \quad (4)$$

and

$$\delta_{jk} = g_{\zeta}(\lambda_{jk}), \quad (5)$$

and where $g_{\zeta}(\lambda_{jk})$ is a deterministic polynomial function parameterized by ζ (shared by all cells and genes) describing the relationship between transcript proportion λ_{jk} and the dispersion δ_{jk} . Details on the estimation of ζ can be found in [Supplementary Methods 1](#).

Prior knowledge of gene parameters

Prior knowledge about λ_{jk} is specified as a transformed Beta distribution:

$$P(\lambda_{jk}) = \text{Beta}(\lambda_{jk}^{(\alpha)}, \lambda_{jk}^{(\beta)}) \times (\lambda_j^{(\max)} - \lambda_j^{(\min)}) + \lambda_j^{(\min)} \quad (6)$$

where $\lambda_j^{(\min)}$ and $\lambda_j^{(\max)}$ denote the minimum and maximum possible values for λ_{jk} . $\lambda_j^{(\min)}$ and $\lambda_j^{(\max)}$ are set as follows. For each gene we compute the pseudobulk relative proportion (i.e. $\frac{\sum_{i=1}^p X_{ij}}{\sum_{i=1}^p L_i}$). Assuming relatively balanced numbers of cells from each collection time point, the pseudobulk relative proportion is a reasonable estimate of each genes' mesor. Moreover, based on analysis of past scRNA-seq circadian datasets, the largest observed fold changes of cycling genes from mesor to peak are around 10-fold (i.e. 1 in $\log_{10}$ space), while most have fold changes of around 2-fold. As such, we define $\lambda_j^{(\min)}$ and $\lambda_j^{(\max)}$ to account for fold changes of around 30-fold to be conservative (1.5 in $\log_{10}$ space):

$$\log_{10}(\lambda_j^{(\min)}) = \log_{10}\left(\frac{\sum_{i=1}^p X_{ij}}{\sum_{i=1}^p L_i}\right) - 1.5 \quad (7)$$

$$\log_{10}(\lambda_j^{(\max)}) = \log_{10}\left(\frac{\sum_{i=1}^p X_{ij}}{\sum_{i=1}^p L_i}\right) + 1.5 \quad (8)$$

By default, **Tempo** sets $\lambda_{jk}^{(\alpha)} = \lambda_{jk}^{(\beta)} = 1$, which assumes a non-informative prior over the domain of possible λ_{jk} values.

Approximate posterior inference

Using our prior knowledge of the gene parameters and the observed data, we seek the following joint posterior distribution of our gene parameters:

$$P(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta}) \sim P(\mathbf{X} \mid \boldsymbol{\lambda}, \boldsymbol{\theta}) P(\boldsymbol{\lambda} \mid \boldsymbol{\theta}) \quad (9)$$

where $\boldsymbol{\lambda}$ is a $p \times q$ dimensional matrix containing the transcript proportion parameters for each gene at each time point, and

$$P(\mathbf{X} \mid \boldsymbol{\lambda}, \boldsymbol{\theta}) = \prod_{i=1}^n \prod_{j=1}^p \prod_{k=1}^q P(X_{ij} \mid (\Theta_i = k)) \quad (10)$$

and

$$P(\boldsymbol{\lambda} \mid \boldsymbol{\theta}) = \prod_{j=1}^p \prod_{k=1}^q \lambda_{jk} \quad (11)$$

No known analytic solution exists to $P(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta}) \sim P(\mathbf{X} \mid \boldsymbol{\lambda}, \boldsymbol{\theta}) P(\boldsymbol{\lambda} \mid \boldsymbol{\theta})$. Moreover, asymptotically exact estimation approaches, such as Markov-chain Monte Carlo and full grid sampling, do not scale well to droplet-based scRNA-seq datasets that can contain thousands (and sometimes tens of thousands) of cells.

For a computationally efficient solution to estimate $P(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$, we use variational inference, an optimization-based approximate Bayesian inference approach.

In brief, we initialize an approximate posterior distribution $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$ that follows a transformed Beta distribution with differentiable parameters describing its shape. $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$ is then optimized to minimize its KL divergence with $P(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$ by minimizing the evidence lower bound (ELBO) objective function. Additional details of the initialization, generative process, and optimization procedure of $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$ can be viewed in [Supplementary Methods 2](#).

Conversion of Bayesian ANOVA parameter posteriors to circadian cosinor parameter posteriors

Given our approximate posterior $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$, for gene j we can generate a sampled waveform $\lambda_j^* = (\lambda_{j1}^*, \lambda_{j2}^*, \lambda_{j3}^*, \lambda_{j4}^*)$, where λ_{j1}^* , λ_{j2}^* , λ_{j3}^* , and λ_{j4}^* correspond to sampled gene proportions from ZT0, ZT6, ZT12, and ZT18, respectively.

Using this sampled waveform, we apply a Fast Fourier Transform, which yields sampled cosinor parameters corresponding to 12-hour and 24-hour components. We denote the sampled 24-hour circadian components for gene j as μ_j^* , A_j^* , and ϕ_j^* , corresponding to the mesor, amplitude, and acrophase, respectively. Sampling from $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$ repeatedly, we can generate samples of μ_j , A_j , and ϕ_j to approximate their posterior distributions.

Identification of confident circadian cycling genes

Confident circadian cyclers were identified using three steps. First, we computed the Bayes factor of each gene using $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$, where 5 λ_j^* samples were drawn for each gene at each time point. Second, we computed the Bayesian evidence of each gene using an altered version of $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$, where the 24-hour circadian component was subtracted out. To do this, 5 $\lambda_j^{*(\text{flat})}$ samples were drawn for each gene. Samples were then converted to the frequency domain using the Fast Fourier Transform, and the 24-hour circadian component amplitudes were set to zero. Frequency domain samples were then transformed back to the original domain using the Inverse Fast Fourier Transform, and then the Bayesian evidence was computed. Third, we compute a Bayes factor for each gene equal to the Bayesian evidence associated with $q(\boldsymbol{\lambda} \mid \mathbf{X}, \boldsymbol{\theta})$ divided by the Bayesian evidence associated with the 24-hour component subtracted posterior. We refer to this Bayes factor as the circadian Bayes factor. Confident cycling genes were defined as those with a circadian Bayes factor greater than or equal to 100, unless otherwise stated.

Supplementary Results

Bmal1 knockout effects on the male aorta circadian transcriptome

As anticipated, male $Bmal1^{fl/fl};CAGG^{creERt2/+}$ mice showed decreased rhythmic circadian transcription compared to C57BL/6J mice. Whereas hundreds of genes show strong circadian cycling in male C57BL/6J aortic cell types, only tens of such cyclers were detected in $Bmal1^{fl/fl};CAGG^{creERt2/+}$ mice (Supplementary Figure 7a). Genes cycling in either condition show substantially reduced amplitudes in $Bmal1^{fl/fl};CAGG^{creERt2/+}$ mice (Supplementary Figure 7b). Moreover, as expected, core clock gene expression waveforms are nearly flat (Supplementary Figure 7c).

Several aspects suggested inducible Bmal1 KO affected genes implicated in synthetic phenotypes in SMCs. The master contractile regulator Myocd was downregulated at ZT6, while contractile genes Myh11, Tpm1, and Smtn were downregulated at all time points (Supplementary Figure 8a). This was echoed by an enrichment for genes involved in smooth muscle cell contraction among genes with decreased circadian mesors. Klf4, whose activity has been documented to promote SMC phenotypic switching and repress myocardin-dependent contractile expression [39], was upregulated at all time points but ZT6, the effects of which were more modest in the descending thoracic aorta. Genes involved in apoptosis, p53 signaling, and proteoglycans in cancer were enriched for increased mesor. Several factors in the TGF- β pathway activation were perturbed, with Tgfr1 downregulated at all time points, Tgfr2 downregulated at ZT12 (Supplementary Figure 8b), and genes with Smad4 promoter motifs enriched for downregulated genes, particularly at ZT12. However, effects on TGF- β activation were nuanced, as inhibitors of TGF- β signaling Smad6, Smad7, Smad9, and Lefty1 were all downregulated at ZT12. Moreover, TGF- β regulators Bmp4 and Bmp6 were downregulated at ZT0, while Bmp3 was down at all time points and lost cycling. Loss of Bmp has previously been observed in phenotypic switching [77]. Functional assessments

of alterations in TGF- β signaling activation are warranted. Genes involved in focal adhesion, adherens junctions, and tight junctions were notably enriched among genes with downregulated mesors, suggesting SMCs may lose sources of mechanical stimuli (from both extracellular matrix and adjacent cells) essential for maintaining the contractile state [78, 79]. Hk2 lost cycling and its mesor was decreased, indicating decreased glycolytic metabolism. Moreover, genes involved in choline metabolism were enriched for downregulation at ZT12, which has previously been implicated in inhibiting SMC phenotypic switching [80].

In addition to alterations in contractile, TGF- β , mechanical sensing, and metabolic expression, proliferation increases are common to synthetic SMC phenotypes. However, our data suggest inducible global Bmal1 KO may repress SMC proliferation. Wee1, a clock-controlled inhibitor of the cell cycle G2 to M transition, oscillates in all cell types and peaks around ZT12 in Bmal1^{+/+} mice. In Bmal1^{fl/fl}:CAGG^{creERT2/+} male mice, Wee1 expression loses oscillation and is constitutively lowly expressed (Supplementary Figure 8c). In contrast, Cdkn1a, a clock-controlled regulator of the cell cycle G1 arrest in response to DNA damage [81], oscillates in all cell types and peaks around ZT0 in Bmal1^{+/+} mice. In Bmal1^{fl/fl}:CAGG^{creERT2/+} mice, Cdkn1a expression loses oscillation and is constitutively highly expressed (Supplementary Figure 8c). Existing literature [82] suggests Cdkn1a activation can inhibit S-phase DNA damage induced by Wee1 inhibition. As such, Cdkn1a expression may increase in response to Wee1 expression loss in Bmal1^{fl/fl}:CAGG^{creERT2/+} mice to preserve DNA integrity. However, while Cdkn1a activity is presumed to inhibit cell cycle progression, its effects may be context dependent. Previous groups demonstrate Cdkn1a overexpression stimulates increased proliferation in SMCs and apoptosis inhibition [83]. Altogether, while Wee1 and Cdkn1a effects suggest altered SMC proliferation after Bmal1 KO, the directionality is unclear and should be the subject of future investigation.

Beyond effects on genes relevant to SMC phenotypic switching, inducible Bmal1 KO has a profound effect on sex hormone dependent gene expression. In particular, we observed several

similarities between male Bmal1 KO SMCs and SMCs of female WT mice. Similar to female Bmal1 WT SMCs, Bmal1 KO SMCs downregulate Hspa1a and Hspa1b, particularly at ZT0 and ZT6 (Supplementary Figure 8d). Atf3, Nr4a1, and Nr4a3 were all downregulated at ZT6 and cycled strongly in male KO SMCs relative to WT. Previous observations made in mouse liver suggest conventional global Bmal1 KO leads to “feminized” expression and increased estrogen receptor activity [84]. Our data also reflect this finding. While moderately expressed in the descending aorta and lowly expressed in the ascending aorta with flat expression in WT mice, Esr1 is highly upregulated at all time points after inducible Bmal1 KO (though notably more highly expressed in the descending aorta). This suggests clock mechanisms may repress its expression and may mediate the cardioprotective effects observed by Yang et al. [85].

We observed large alterations in the expression of matrix metalloproteases and metalloprotease inhibitors in SMCs. Mmp14 was increased at ZT12/18 in the ascending aorta/arch, while Mmp2 was increased at all time points in all regions of the aorta. While this may suggest increased rates of extracellular matrix degradation, Timp4, an inhibitor of metalloprotease activity, was strongly upregulated at all time points, particularly in the ascending aorta/arch.

Several effects were common to multiple cell types. In particular, decreases in focal adhesion were common to SMCs, fibroblasts, and ECs, suggesting clock function in the endogenous vascular cell types may be essential for proper mechanical sensing of extracellular matrix. Aldosterone synthesis was downregulated in SMCs and fibroblasts. Complement activation was upregulated in SMCs and fibroblasts, which has been suggested to be atherogenic [87, 88]. C3, C1s1, and C4b were upregulated at all time points in fibroblasts (Supplementary Figure 8e), while C1s1 and C4b were upregulated at all time points in SMCs. C3 notably gained cycling, with increased expression at ZT6 and decreased expression at ZT18, relative to WT SMCs. This gain of SMC C3 cycling may be due to gain of cycling in Atf3, which has been

previously described to inhibit C3 transcription and cycles in antiphase in KO SMCs [59].

Our results suggest Bmal1 KO may affect macrophage recruitment. While myeloid cells are rhythmically recruited to the vasculature via Ccl2-Ccr2 signaling [4], with Ccr2 macrophage expression peaking at ZT0 in C57BL/6J mice, rhythmic expression of Ccr2 was delayed nearly 12 hours and its mesor was modestly increased in Bmal1^{fl/fl}:CAGG^{creERT2/+} macrophages (Supplementary Figure 8f). Moreover, rhythmic signaling via the Cx3cl1-Cx3cr1 axis [89, 90, 91] appeared disrupted. Cx3cl1 and Cx3cr1 expression cycled and peaked around ZT0 in WT ECs and macrophages, respectively (Supplementary Figures 8f and 8g). After Bmal1 KO, both genes were flat and constitutively highly expressed. Altogether, this suggests loss of Bmal1 expression may drive chronic macrophage recruitment to the aorta.

Bmal1 KO may promote a more proinflammatory environment. In ECs, expression of the anti-inflammatory signaling gene Cxcl12 was greatly decreased at ZT0, and expression of pro-inflammatory signaling gene Cd74 was increased at ZT12. In macrophages, expression of pro-inflammatory genes Cd72 and Il1b were greatly increased at ZT18, while expression of anti-inflammatory genes Cd200 and Ccl24 was greatly decreased. While Mmp9 cycles strongly in WT male macrophages, peaking at ZT6, it is constitutively highly expressed after Bmal1 KO, suggesting KO macrophages degrade extracellular matrix and promote angiogenesis. This is likely due to the loss of Nr1d1 expression [92]. Genes increasing in mesor in macrophages were further enriched for positive regulation of NF κ b activity. Altogether, these results suggest that the Bmal1 KO in male mice may bias macrophages to express genes associated with a proinflammatory state, as has been previously described [92].

Effect of sex on Bmal1 knockout effect

KO effects in cell types are strongly concordant between male and female cell types (Supplementary Figure 9). In both sexes, the number of cell-type-specific confident circadian cyclers was decreased. However, females retained higher numbers of rhythmic gene sets

(Supplementary Figure 10a).

Among unique confident cyclers in female SMCs, dusk cyclers were enriched for TNF- α signaling and inhibition of MAPK signaling, while dawn cyclers were enriched for ECM organization and Pi3k-Akt signaling. Moreover, estrogen signaling genes were enriched at dusk, particularly in SMCs of the ascending aorta/aortic arch. Similar to its expression in male SMCs, *Esr1*'s mesor greatly increases in female SMCs after *Bmal1* KO. Unlike in males, *Esr1* cycles strongly in female SMCs, peaking at ZT15 (Supplementary Figure 11a). Several other components of the estrogen signaling pathway uniquely cycled in female KOs, such as *Hsp90aa1*, *Fkbp5*, *Hbegf*, *Atf4*, peaking at dusk. Among estrogen signaling cycling genes induced by KO were *Jun* and *Fos*, which gained expression at ZT12, specifically in the ascending aorta/aortic arch of female SMCs. This suggests increased Ap1 transcription factor activity at ZT12, which has been implicated in SMC proliferation and neointimal formation [93, 94, 95]. Interestingly, *Bmal1* KO in female SMCs induces an acrophase shift of *Atf3* to ZT12. As *Atf3* may antagonize Ap1 activity [96], this may reflect a compensatory effect to buffer against changes in Ap1 activity after KO.

One of the most striking sex differences in *Bmal1* WT SMCs was in the expression of *Hspa1a* and *Hspa1b* (Supplementary Figure 11a). Both were among the strongest cyclers in female SMCs, peaking around ZT21, with considerably lower expression in females at ZT12. However, after *Bmal1* KO, *Hspa1a* and *Hspa1b* appeared similar between the sexes, cycling in both sexes with an acrophase around ZT15 and modestly lower expression in females during the light phase. Previous reports suggest *Bmal1* KO leads to “feminized” circadian expression liver, and that these effects are likely due to loss of central oscillators [84]. Our data suggest expression of WT confident cyclers in SMCs becomes more similar between sexes after KO (Supplementary Figures 11b and 11c). This suggests that sex-specific expression may be mediated, in large part, through expression of circadian cycling genes.

Several aspects suggest female SMCs may proliferate less than male SMCs after *Bmal1* KO.

Mcm6, a mini-chromosome maintenance protein required for replication, cycles in female WT SMCs, with an acrophase around ZT15. In female SMCs, Mcm6 expression is downregulated at all time points, an effect not observed in males (Supplementary Figure 11a). Plat, a plasminogen activator that has previously been shown to be a mitogen for SMCs [97], loses expression uniquely in females at ZT12 and ZT18 (Supplementary Figure 11a). This effect is more pronounced in SMCs of the ascending aorta/aortic arch. Similarly, Apoe, which can inhibit SMC proliferation by activating inducible nitric oxide synthase [98], gains expression more strongly in females at ZT12 and ZT18 than in males (Supplementary Figure 11a), which is particularly pronounced in the ascending aorta/aortic arch. This same trend was observed for Errf1, which inhibits neointimal formation [99]. Nonetheless, Rgs5, which inhibits SMC proliferation [31, 100, 101] by antagonizing G-coupled protein receptor signaling, demonstrates decreased expression at ZT6 in the ascending aorta/aortic arch and at all time points in the descending thoracic aorta. In the absence of G-coupled protein receptor signaling, the net effect may still be an inhibition of SMC proliferation in female KO SMCs. Nonetheless, understanding how these effects on key proliferation genes ultimately affects proliferation will require experimental follow up.

Female SMCs may also be less likely to migrate after Bmal1 KO. Serpine1 (a.k.a. Pai1), which can stimulate SMC migration and neointimal formation [102], is downregulated at ZT0 in the ascending aorta/aortic arch of females relative to their male KO counterparts (Supplementary Figure 12a); Ace, an angiotensin converting enzyme promoting SMC migration [103], is only downregulated in female SMCs of the aortic arch/ascending aorta at ZT12/18 (Supplementary Figure 12a). Female SMCs may also be more predisposed to oxidative stress after KO. Nupr1, a key gene in antioxidant activity [104], is specifically downregulated in female SMCs of the aortic arch/ascending aorta at ZT12/18.

While globally upregulated in male SMCs after Bmal1 KO, Cdkn1a expression was not altered by Bmal1 KO in male macrophages. However, macrophage Cdkn1a expression was

strongly downregulated at ZT6 in female macrophages, suggesting an increased likelihood to progress through the cell cycle at these time points (Supplementary Figure 12b). Several KO effects observed in male macrophages, were shared by females, such as an increased mesor for *Cd72* and *Il1b*. However, both genes displayed acrophase shifts. Similarly, *Ccr2* peaked around ZT15 in male macrophages after *Bmal1* KO, while *Ccr2* peaks around ZT20 in female macrophages, suggesting sex dependent differences in myeloid recruitment after clock KO (Supplementary Figure 12b). Genes with mesor decreases in female KO relative male KO macrophages were enriched for lipopolysaccharide response genes and cellular response to cytokine stimulus, such as *Zfp36*, *Ccl3*, *Tnfaip3*, *Ccl2*, *Cd14*, *Tnf*, and *Ccl4*, with lower expression in females KO macrophages around ZT6–ZT12. Moreover, genes downregulated at ZT6 in female KO macrophages relative to male KO macrophages were enriched for NF- κ B signaling. Altogether, these results support that *Bmal1* KO biases female macrophages toward an anti-inflammatory phenotype.

Sex effects on misalignment

Similar to the observations made in the aligned mice, after acute misalignment, females had more confident cycling genes (Supplementary Figure 13a). This finding held after adjusting for discrepancies in cell counts and library sizes (Supplementary Figures 5 and 14). Confident cycling genes common to both males and females demonstrate larger amplitudes (Supplementary Figure 13b). We next evaluated whether female C57BL/6J cell types more quickly entrain to their new LD cycle. To inspect this, we restricted our analysis to confident cycling genes common to the male aligned, male misaligned, female aligned, and female misaligned conditions within SMC subtypes and fibroblasts. For each gene in each sex and cell type, we computed the posterior distribution of the absolute value of the difference between the misaligned and aligned peak times (i.e. acrophase difference distributions). Smaller values indicate better entrainment (i.e. after misalignment, gene exhibits more similar peak time to the aligned peak time). We then asked if we sampled from both the male and female

acrophase difference distributions, what fraction of the time would the samples be smaller in females. Females demonstrated a consistent, albeit modest, improvement in cell type entrainment (Supplementary Figure 15) for the SMC subtypes and fibroblasts.

After acute misalignment with the LD cycle, female SMCs demonstrate transcriptomic changes similar to male SMCs, regardless of expression differences in the aligned condition. For example, *Uba52* and *Mcm6* cycle in aligned female SMCs and weak cycling in male SMCs, with downregulated expression in aligned female SMCs at ZT6 relative to male SMCs. However, after misalignment, *Uba52* and *Mcm6* lose cycling and gain mesor, with expression at similar levels in male and female SMCs.

Several factors implicated in SMC phenotypic switching are altered specifically in female SMCs after acute misalignment. *Id1* and *Id3* expression is lost at ZT12 in all aortic regions in female SMCs, with minimal expression changes in male SMCs (Supplementary Figure 16a). In SMCs, *Id1* and *Id3* are induced by mitogens, regulate SMC proliferation, and may inhibit differentiation [118, 119, 120], suggesting their female-specific perturbation after misalignment may affect synthetic aspects of SMC function. In alignment *Ccn2* – which upregulates *Tgfb β 2* [121], regulates adhesion, and mediates heparin signaling [122], which inhibits SMC proliferation [53] – cycles strongly, peaking around ZT21 in male and female SMCs. After misalignment, *Ccn2* loses cycling more strongly in female SMCs, with considerably less expression in misaligned female SMCs relative to aligned female SMCs at ZT18, particularly in the ascending aorta/aortic arch (Supplementary Figure 16b).

Several aspects of our data suggest endothelial barrier function is decreased in males, but not females, after acute misalignment with the LD cycle. *Egr3*, which is induced after VEGF signaling [123, 124], is downregulated at ZT0/18 in female misaligned ECs relative to male misaligned ECs (Supplementary Figure 17a). *Nrarp*, which is implicated in angiogenesis [125], was down at ZT6 in misaligned females relative to misaligned males (Supplementary Figure 17a). *Fabp4*, inhibition of which decreases VEGF signaling and angiogenesis [126, 127], was

downregulated at ZT0 in misaligned females relative to misaligned males (Supplementary Figure 17a). Altogether, this suggests males may have decreased endothelial barrier after acute misalignment.

After misalignment, female macrophages lose *Dusp2* expression at ZT6/12, a downstream target of NF- κ b [128] and the loss of which has been shown to reduce macrophage secretion of proinflammatory mediators [129]. After misalignment, *Hbegf* expression is lost only in female macrophages. *Hbegf* activity has previously been implicated in promoting macrophage-driven angiogenesis [130, 131] and macrophage-driven SMC proliferation [132]. Moreover, after misalignment, *S100a6* expression is only gained in female macrophages at ZT0. Previous observations suggest *S100a6* activates RAGE signaling [133], which may drive an increase in female macrophage apoptosis at ZT0.

Supplementary Methods 1: Estimating the transcript proportion – dispersion relationship

We model UMI counts as coming from a Negative Binomial (NB) data likelihood model, as it can account for the overdispersion observed in real count data. While many existing applications of NB to scRNA-seq data model dispersions in a gene-specific fashion, these estimates are known to exhibit high amounts of uncertainty. Methods such as *scTransform* [134] and *DESeq2* [135] combat this by an empirical Bayes approach to shrink dispersion estimates, assuming genes with similar means share similar dispersions. We take this a step further and make the simplifying assumption that the transcript proportion–dispersion relationship across all genes is strictly described by a polynomial function $g_{\zeta}(\lambda_{ij})$ parameterized by ζ . The coefficients of this polynomial, ζ , are estimated as follows:

Step 1: Estimate gene-specific proportions and dispersions

We presume gene counts are distributed according to:

$$X_{ij} \sim \text{NB}(\tilde{\lambda}_j \cdot L_i, \tilde{\delta}_j) \quad (24)$$

Where:

$$\mathbb{E}[X_{ij}] = \tilde{\lambda}_j \cdot L_i \quad (25)$$

$$\text{Var}(X_{ij}) = \mathbb{E}[X_{ij}] + \tilde{\delta}_j (\mathbb{E}[X_{ij}])^2 \quad (26)$$

And $\tilde{\lambda}_j$ and $\tilde{\delta}_j$ are the temporary (i.e. only used to estimate ζ) gene-specific transcript proportion and dispersion for gene j , respectively. Under this likelihood model, for each gene we estimate the maximum likelihood transcript proportion $\tilde{\lambda}_j$ and dispersion $\tilde{\delta}_j$.

Step 2: Fit polynomial model describing the transcript proportion – dispersion relationship across all genes

Assume g is a polynomial function whose coefficients ζ defines the transcript log proportion – log dispersion relationship shared across all genes:

$$g_{\zeta}(\lambda_{ij}) = \log(\delta_j) = \sum_{k=1}^{|\zeta|} \zeta_k \log^k(\lambda_{ij}) \quad (27)$$

Using OLS, we estimate the coefficients ζ under the following objective:

$$\arg \min_{\zeta} \sum_{j=1}^p \left(\log(\tilde{\delta}_j) - g_{\zeta}(\tilde{\lambda}_j) \right)^2 \quad (28)$$

where p is the number of genes, and $\tilde{\lambda}_j$ and $\tilde{\delta}_j$ are the temporary gene-specific transcript proportion and dispersion for gene j estimated in Step 1.

Note

We note that dispersion estimates may be upwardly biased as a subset of genes cycling over the circadian cycle will be fit under a mean model that assumes flat expression over the circadian cycle. However, we assume the fraction of cycling genes detected is small, and thus should not upwardly bias the dispersion estimates much.

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Author contributions

Benjamin J. Auerbach (B.J.A.), Soon Yew Tang (S.Y.T), Ronan Lordan (R.L.), Sean T. Anderson (S.T.A.), Mingyao Li (M.L.), and Garret A. FitzGerald (G.A.F.) contributed to the project. The study was conceived of by G.A.F. and B.J.A. S.Y.T., R.L., and S.T.A. performed the mouse experiments and generated the scRNA-seq data from the mouse aorta. B.J.A. performed the data analysis, with input from M.L and G.A.F. B.J.A. wrote the paper with feedback from M.L. and G.A.F.

Competing financial interests

G.A.F is a Senior Advisor to Calico Laboratories. The remaining authors declare no competing financial interests. All authors declare no competing non-financial interests.

Software availability

Scripts used to analyze the data and generate the figures can be found at <https://github.com/bauerbach95/misalignment>.

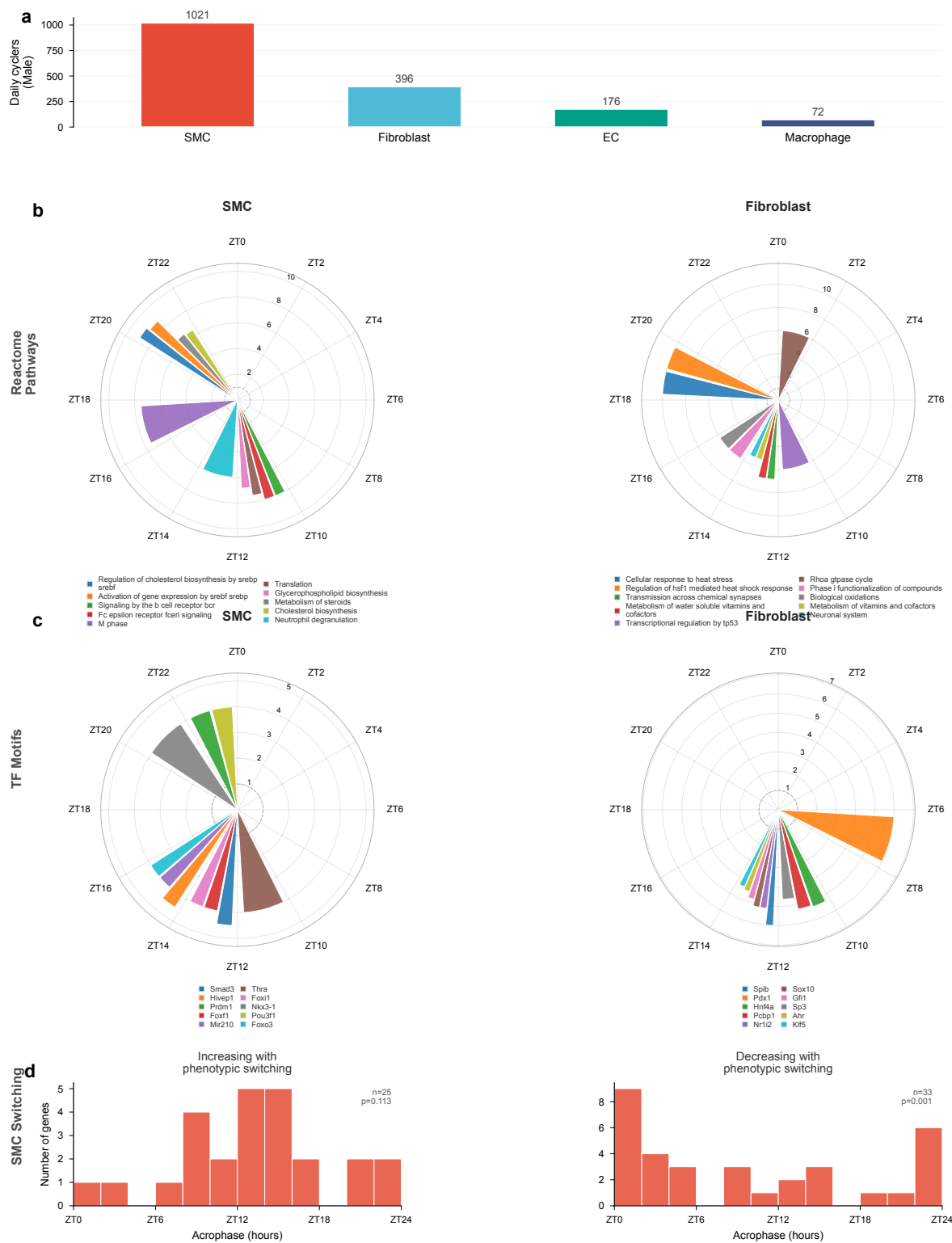
Data availability

Raw sequencing and UMI count matrices were deposited to the Gene Expression Omnibus under accession code **GSEXXXX**.

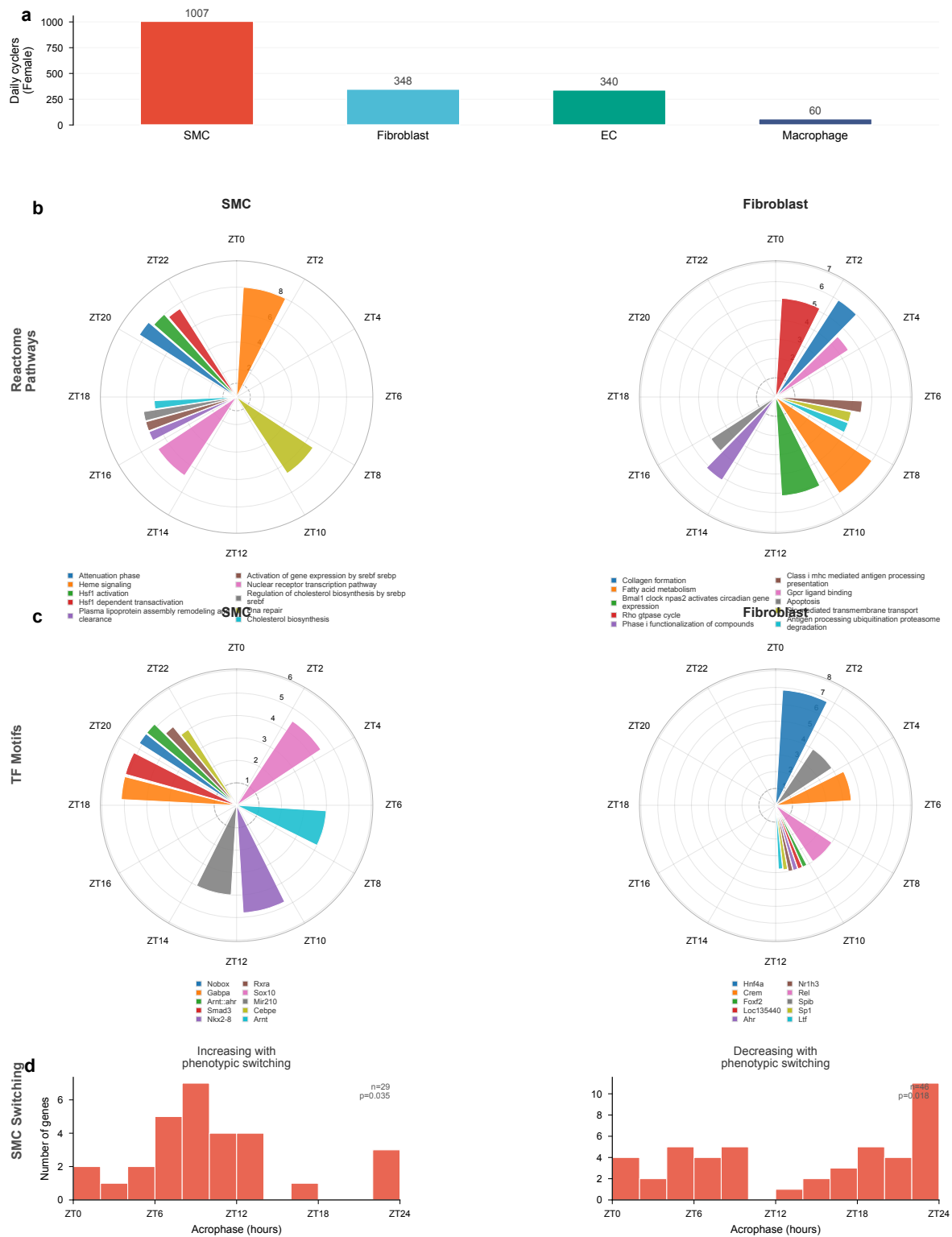
Supplementary Figures

[Supplementary Figure 1 requires assembly from multiple source panels.]

Supplementary Figure 1

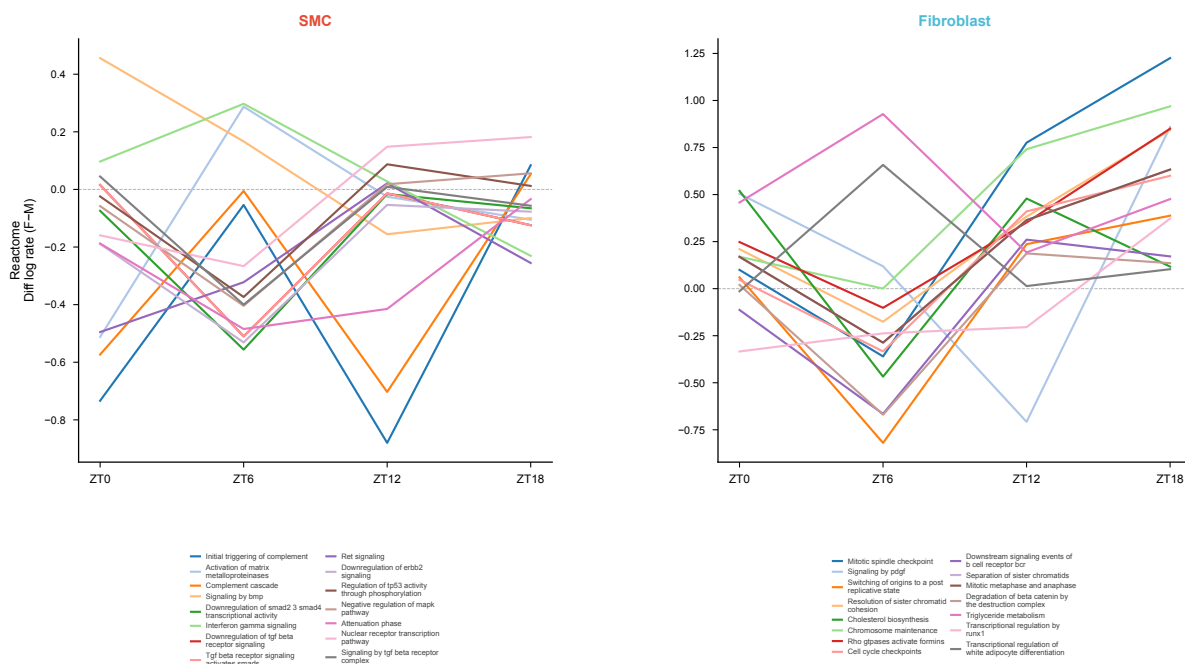


Supplementary Figure 2

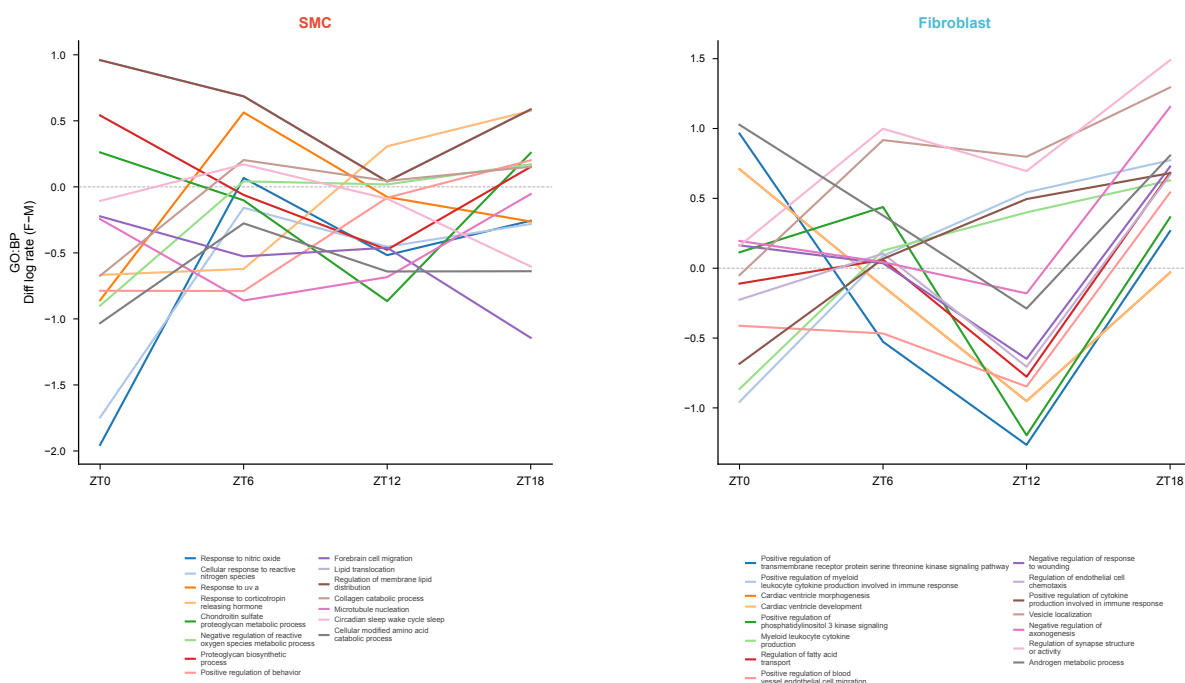


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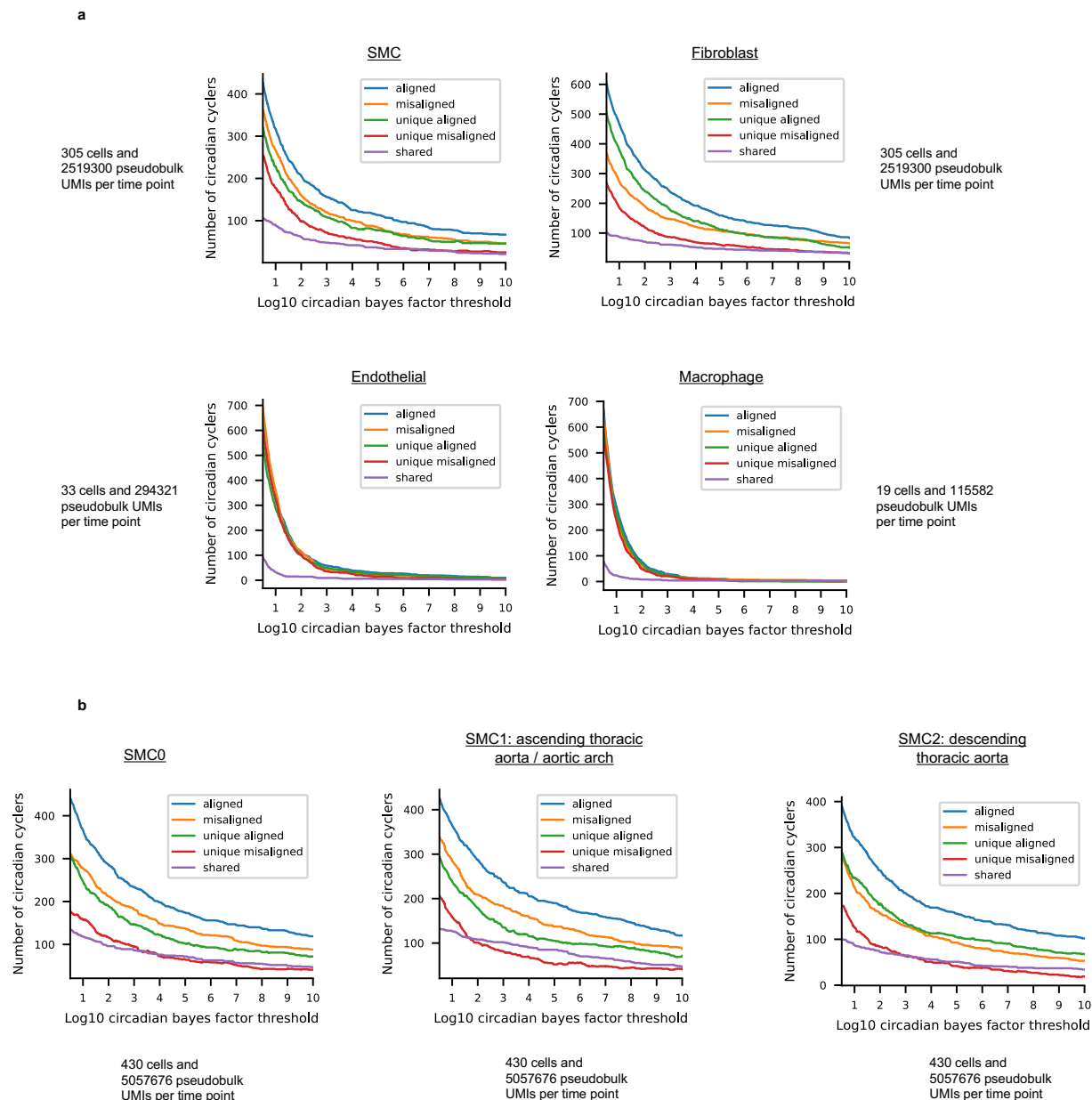
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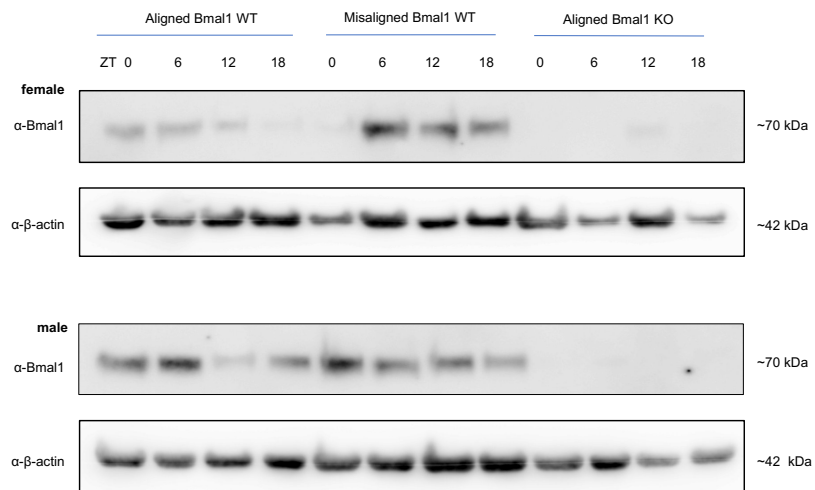
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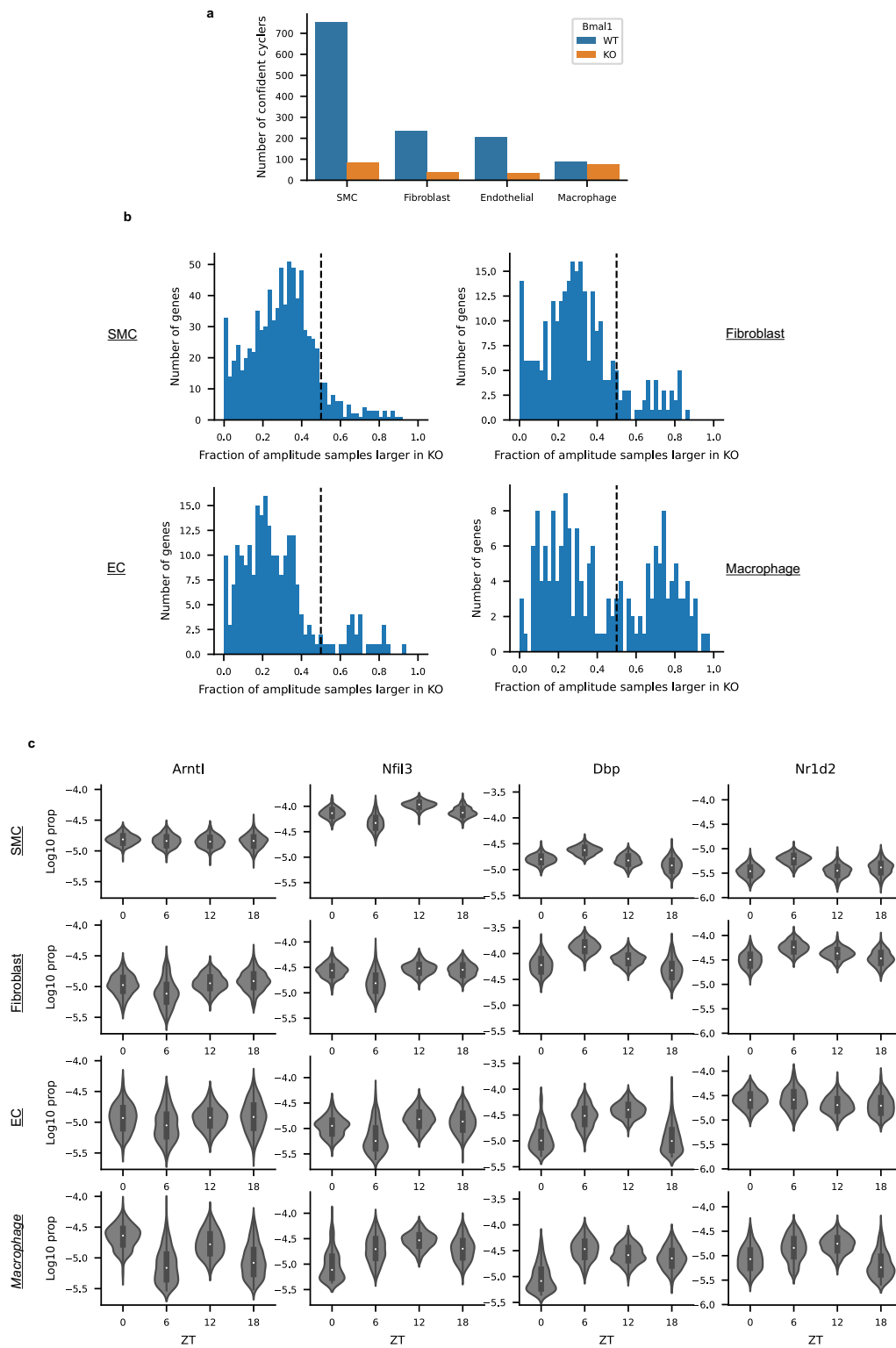
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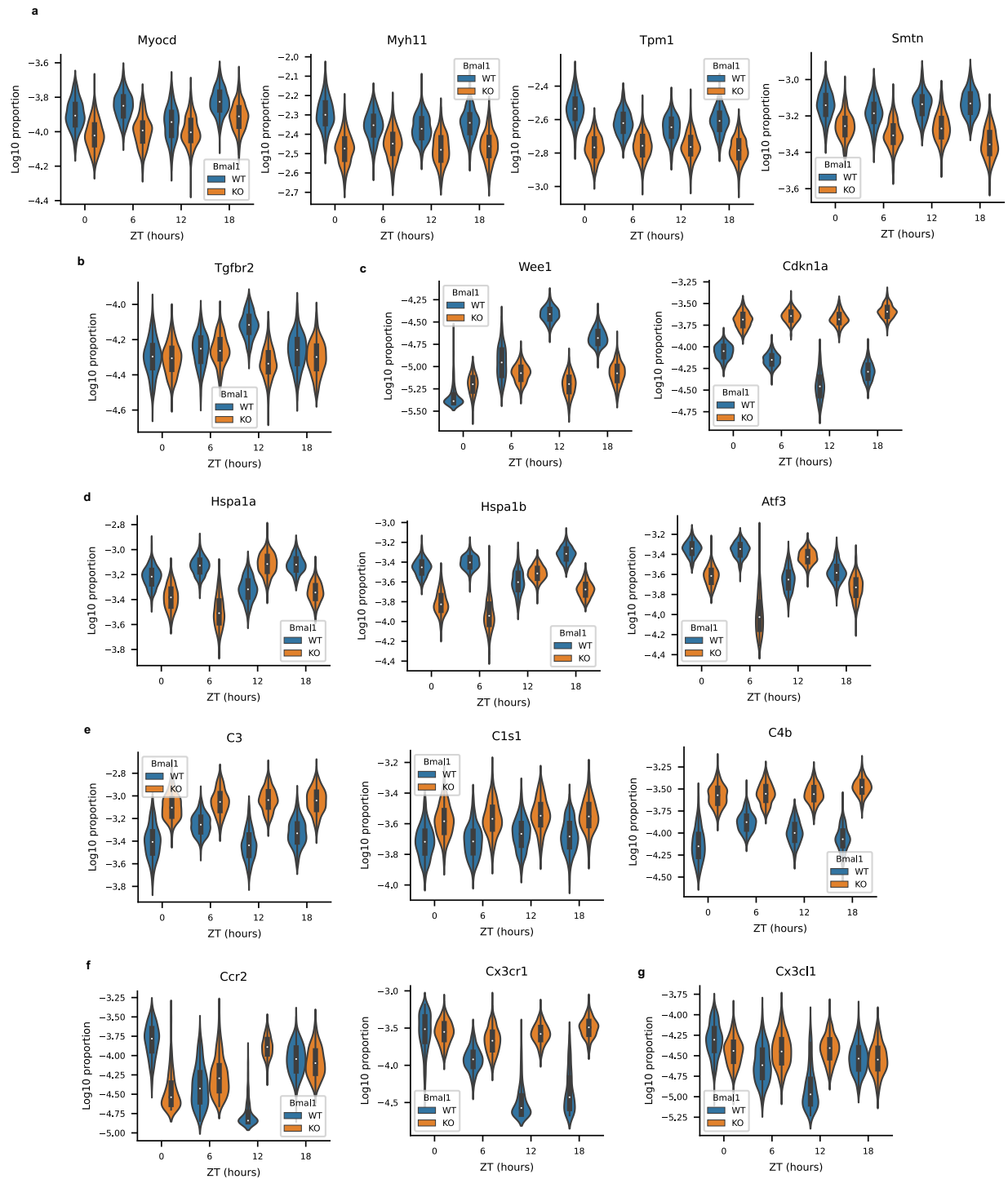
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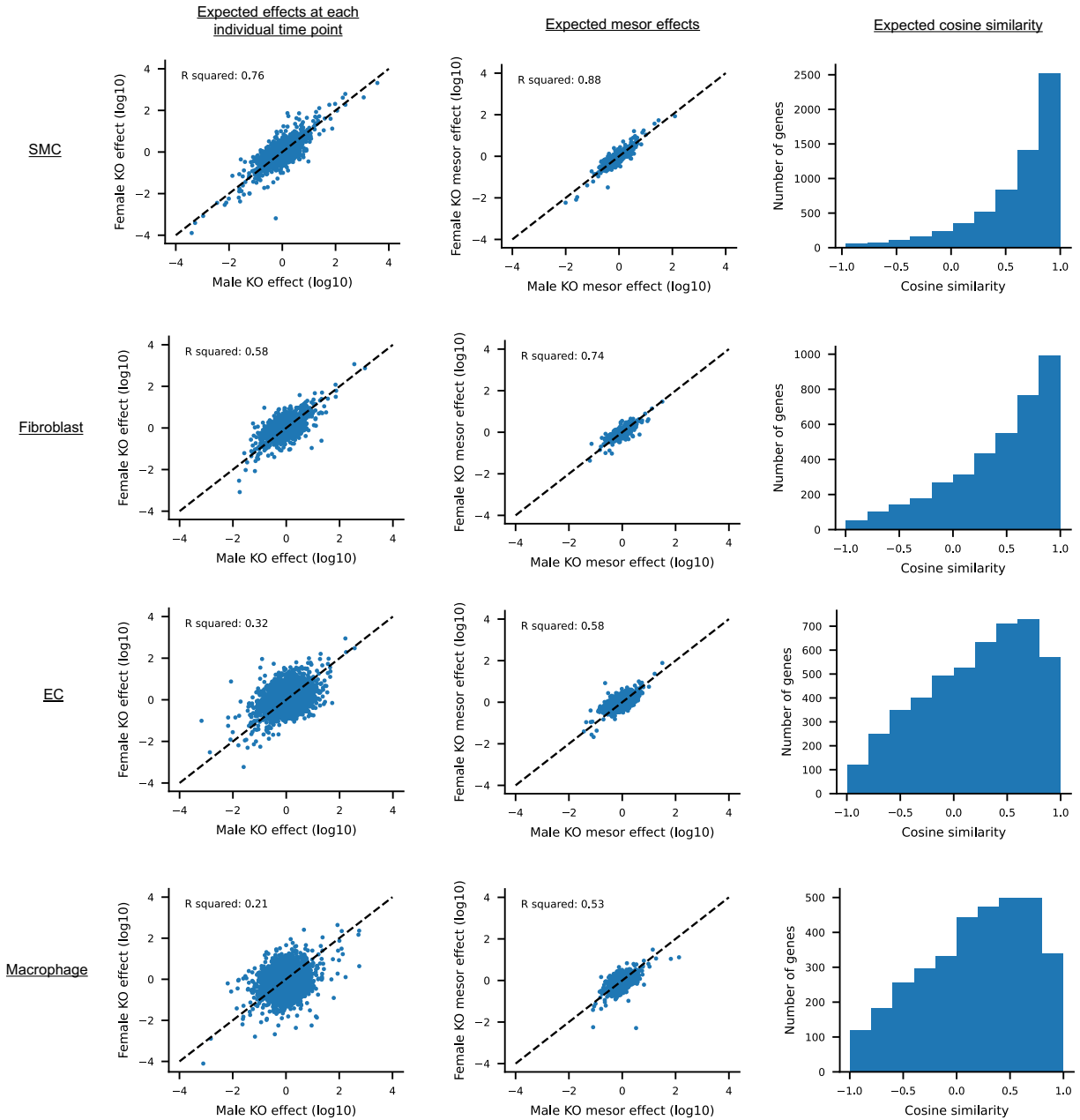
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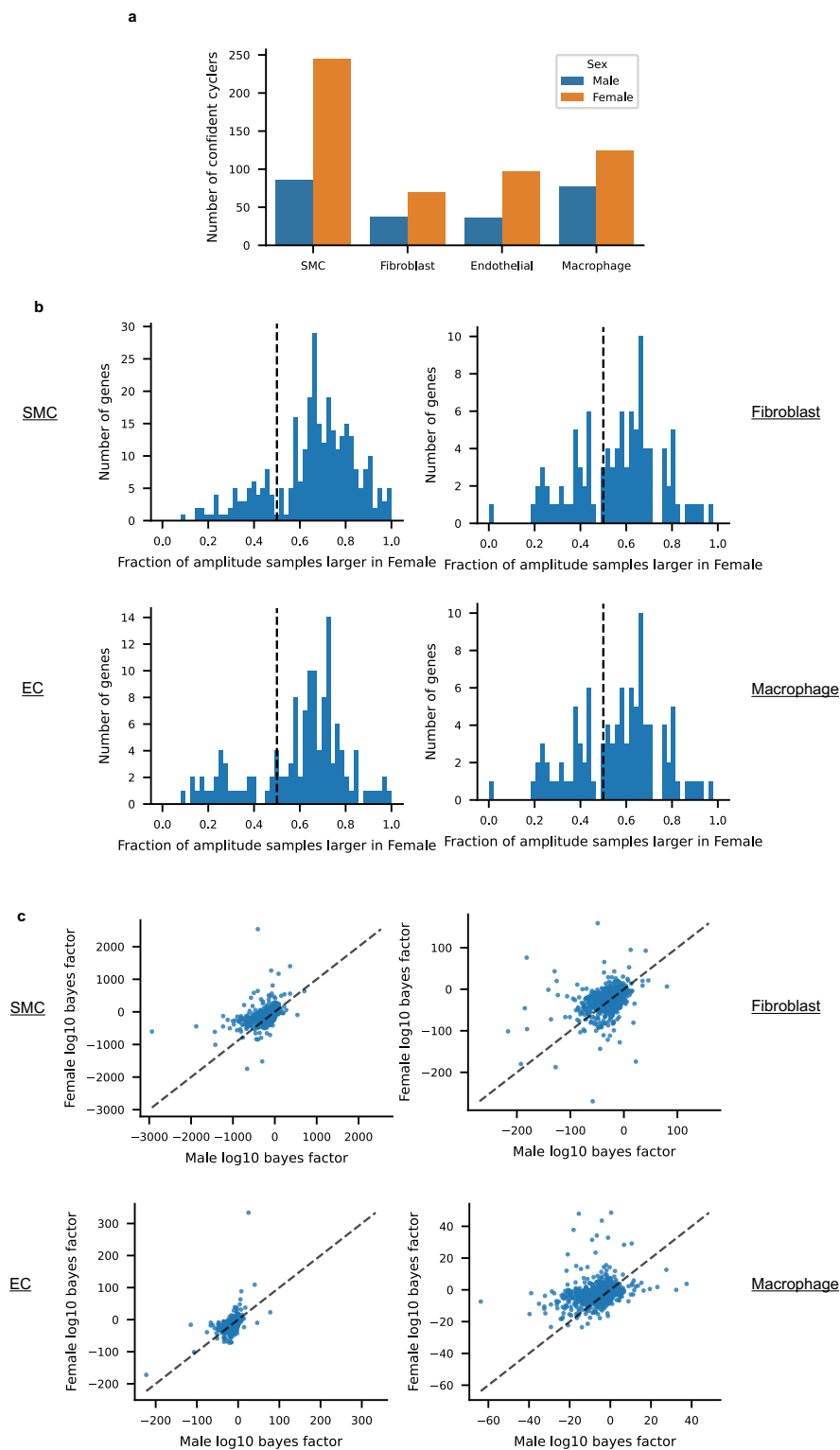
Supplementary Figure 7



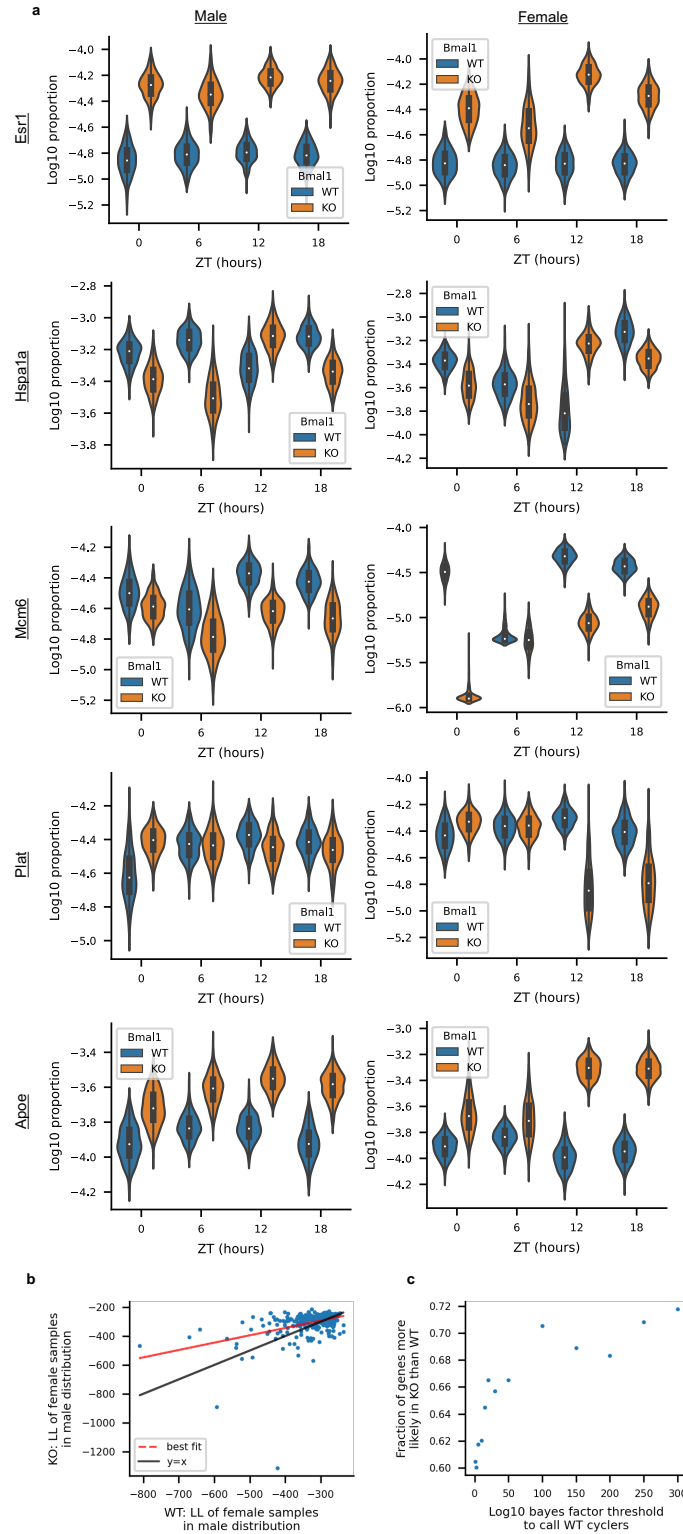
Supplementary Figure 8



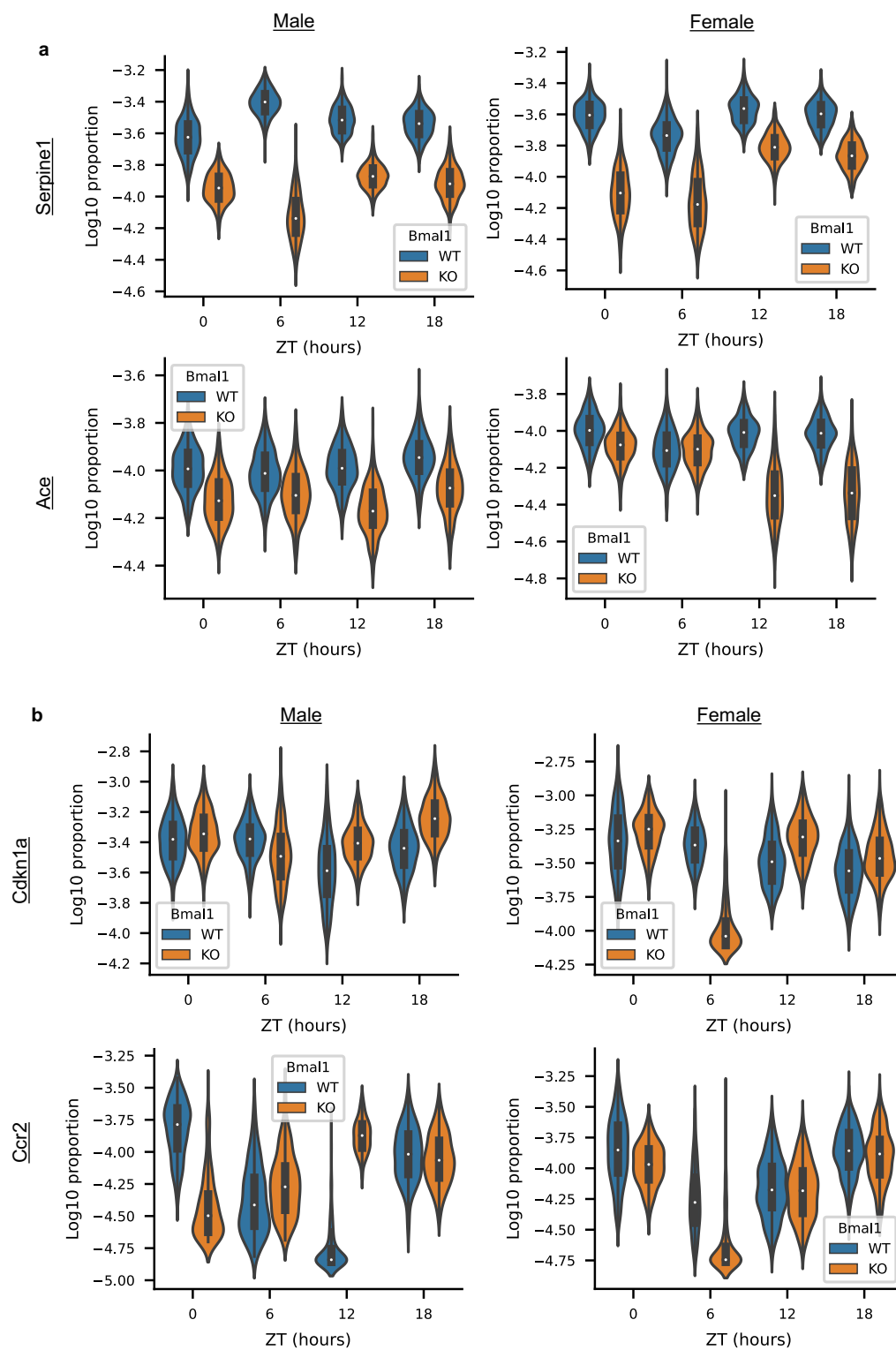
Supplementary Figure 9



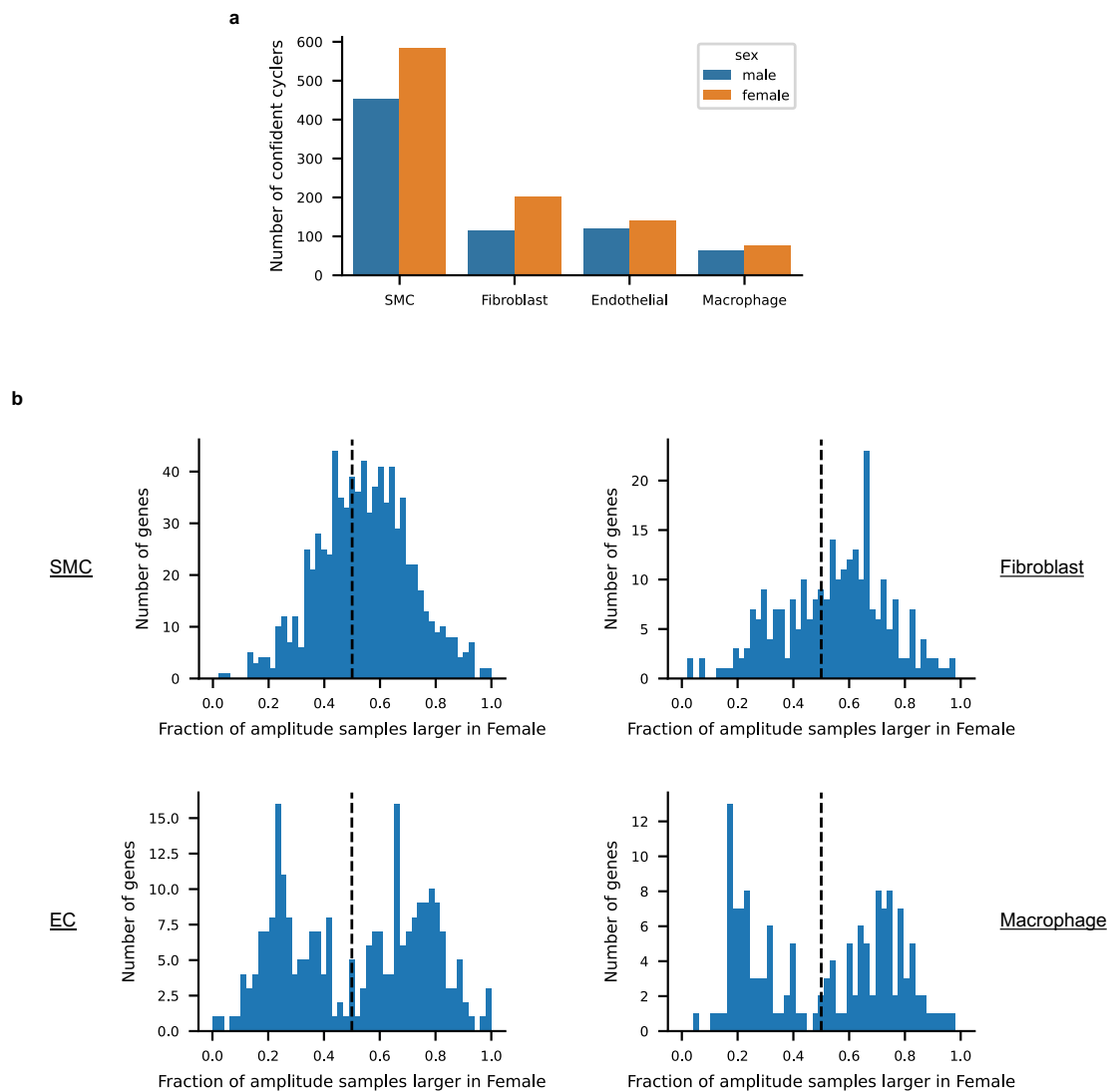
Supplementary Figure 10



Supplementary Figure 11

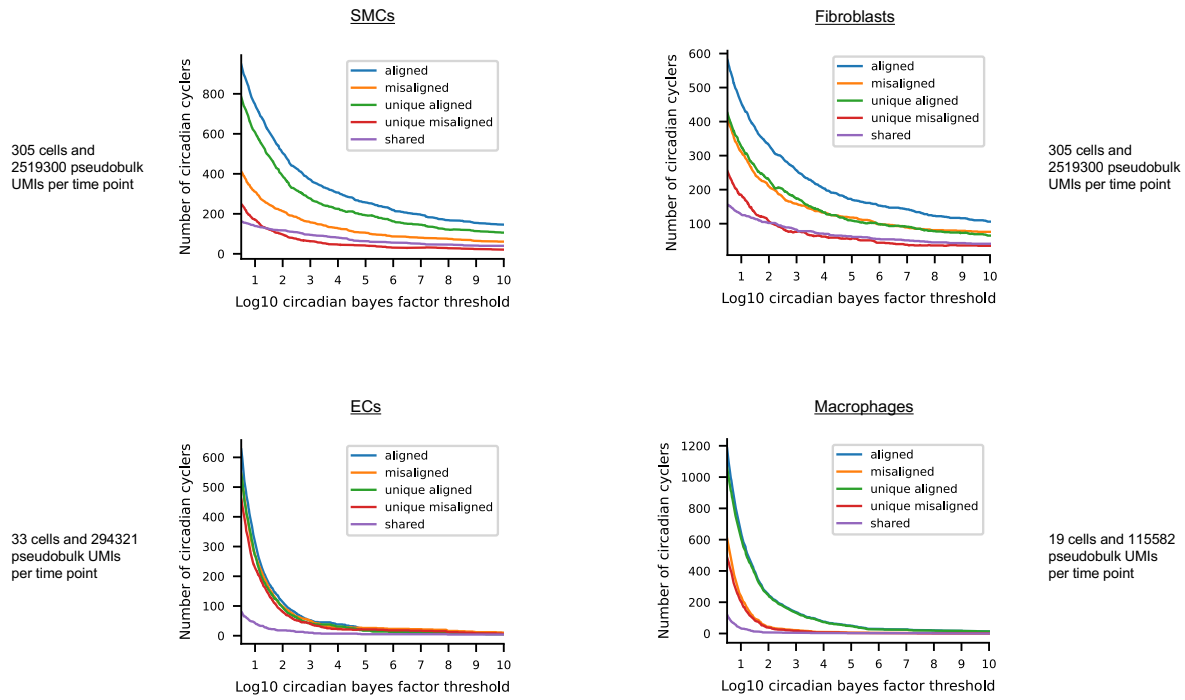


Supplementary Figure 12

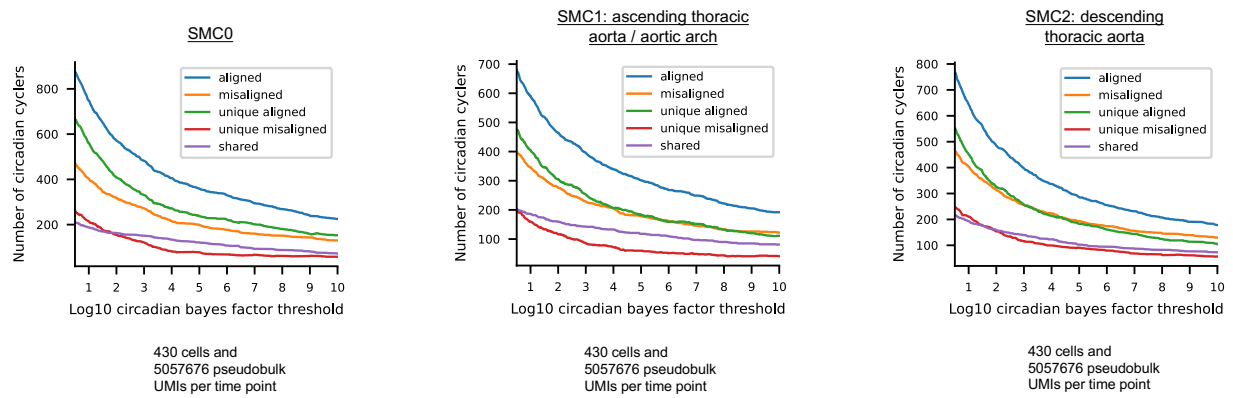


Supplementary Figure 13

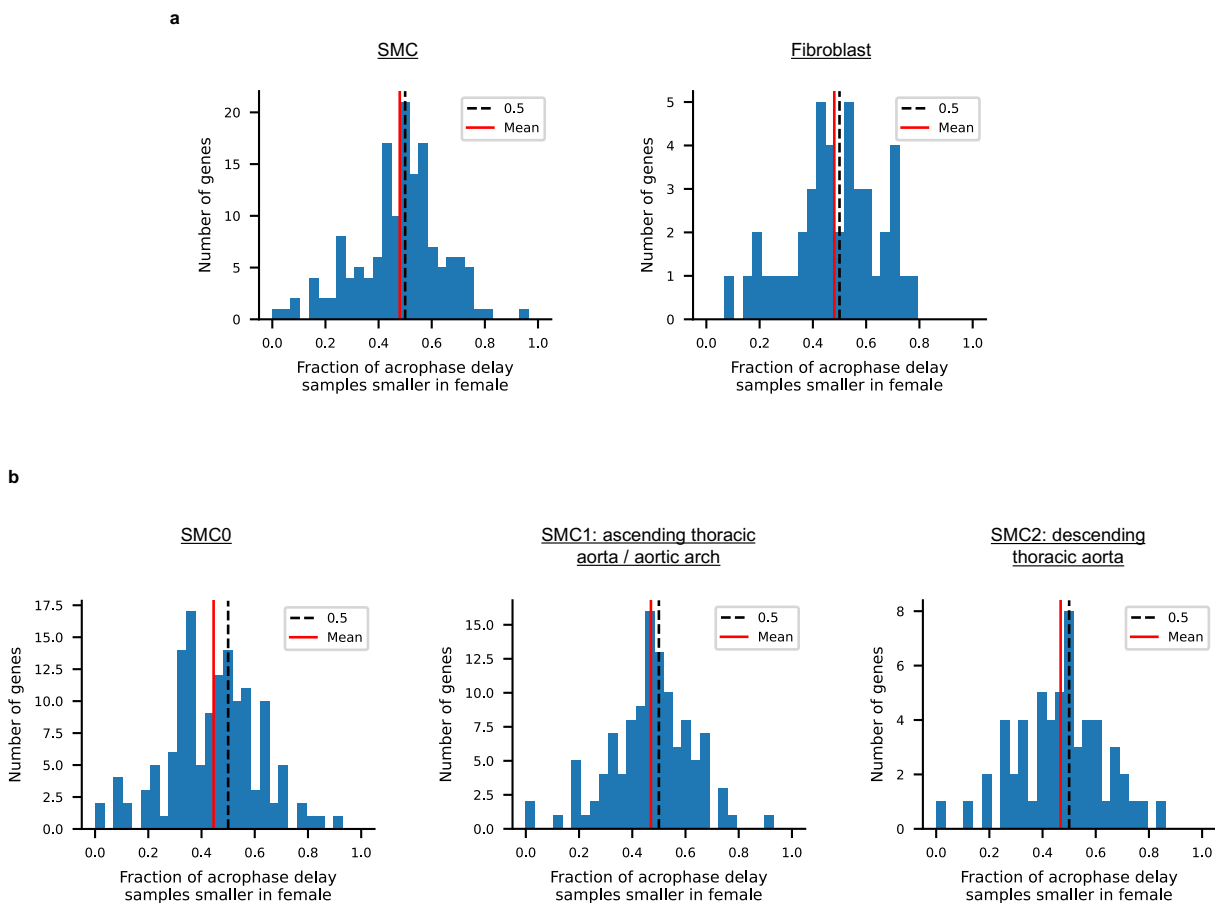
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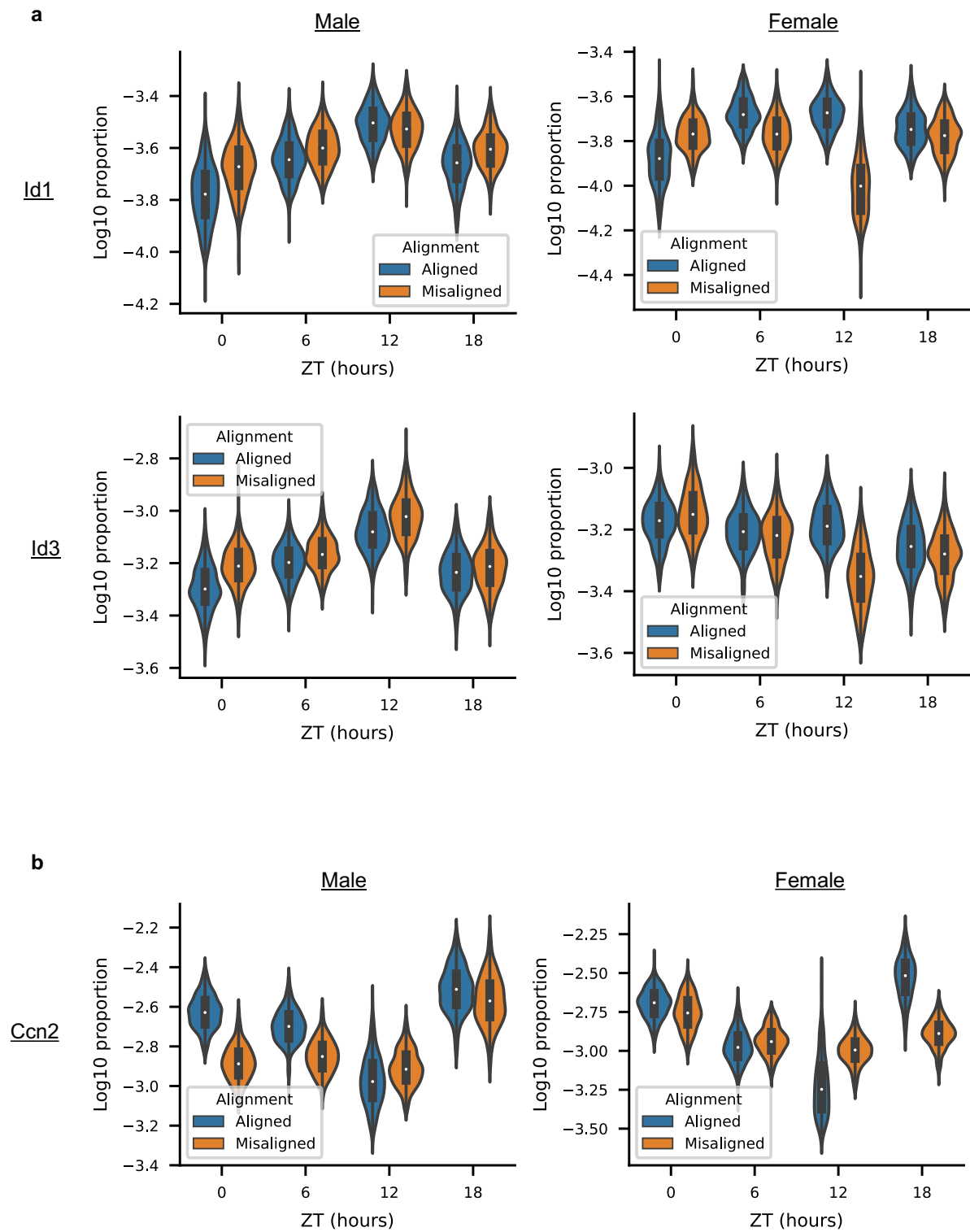
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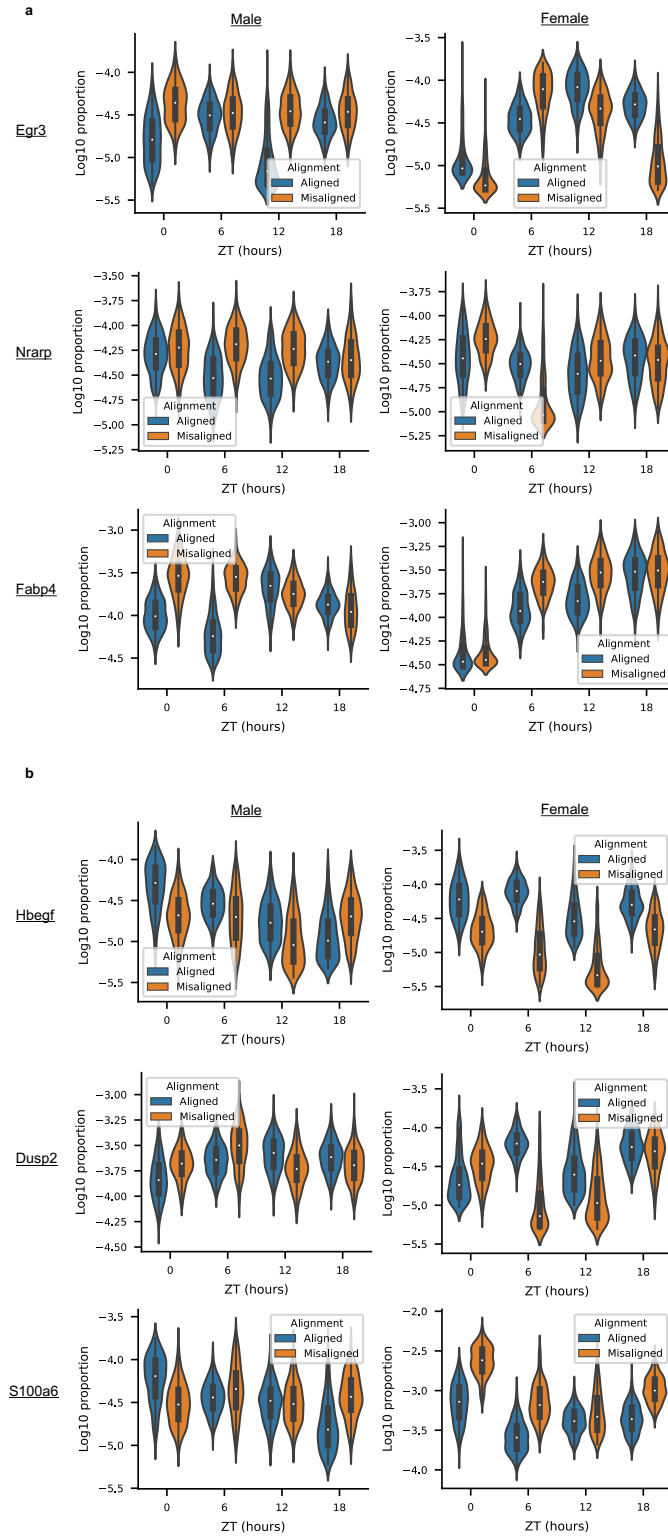
Supplementary Figure 14



Supplementary Figure 15



Supplementary Figure 16



Supplementary Figure 17

Supplementary Figure Legends

Supplementary Figure 1: Validation of circadian cycling detection. a) Comparison of expected acrophases of confident SMC circadian cyclers to their peak times in bulk aorta from Zhang et al. b) Posterior waveforms of core clock genes in major aortic cell types. c) Posterior waveforms of core clock genes in SMC subtypes. d) Histogram of expected acrophases for confidently cycling genes in major aortic cell types.

Supplementary Figure 2: Male-specific daily cycling expression in the mouse aorta. a) Number of daily cycling genes in males for each major cell type. b) Reactome pathway enrichment among male daily cyclers in SMCs and fibroblasts. c) Transcription factor motif enrichment among male daily cyclers in SMCs and fibroblasts. d) Acrophase distributions of male SMC daily cyclers implicated in phenotypic switching, with p -values from window-based permutation tests.

Supplementary Figure 3: Female-specific daily cycling expression in the mouse aorta. a) Number of daily cycling genes in females for each major cell type. b) Reactome pathway enrichment among female daily cyclers in SMCs and fibroblasts. c) Transcription factor motif enrichment among female daily cyclers in SMCs and fibroblasts. d) Acrophase distributions of female SMC daily cyclers implicated in phenotypic switching, with p -values from window-based permutation tests.

Supplementary Figure 4: Extended waveform enrichment for sex-differential expression. Reactome pathway and GO Biological Process waveform enrichment results for sex-differential temporal expression in SMCs and fibroblasts.

Supplementary Figure 5: Misalignment cycling controlled for cell counts and library sizes. Within each cell type and time point, cell counts and pseudobulk UMIs were downsampled

to match between aligned and misaligned male mice. Number of called cyclers across $\log_{10}$ circadian Bayes factor thresholds for major cell types and SMC subtypes.

Supplementary Figure 6: Western blots of Bmal1 protein. Western blots of Bmal1 protein product from aorta across all conditions collected in the study.

Supplementary Figure 7: General characteristics of the effect of inducible Bmal1 deletion on cycling expression in males. a) Number of confident circadian cycling genes in major cell types of aorta from male Bmal1 wild type and knockout mice. b) Distribution of the gene fraction of posterior amplitude samples being larger in male Bmal1 knockout cell types relative to wild type, considering the union of confident cycling genes. c) Posterior $\log_{10}$ proportion estimates for core circadian clock genes at each sampled time point and in each major cell type of the Bmal1 knockout mice.

Supplementary Figure 8: Bmal1 knockout effects on individual gene expression waveforms in male mice. Posterior circadian waveforms in SMCs for a) genes involved in contractile expression, b) TGF- β signaling, c) cell cycle entry, and d) stress response. e) Posterior circadian waveforms in fibroblasts for complement system genes. f) Posterior circadian waveforms in macrophages for receptors involved in vascular recruitment. g) Posterior circadian waveforms in ECs for ligands involved in vascular recruitment.

Supplementary Figure 9: Concordance of Bmal1 knockout effect in males and females. Left: the posterior Bmal1 KO effect on each gene at each time point in males vs. females. Center: the posterior Bmal1 KO effect on each gene's mesor in males vs. females. Right: the expected cosine similarity between the male and female effects at each time point.

Supplementary Figure 10: General characteristics of the sex and inducible Bmal1 deletion effect on cycling expression. a) Number of confident circadian cycling genes in major cell

types of aorta from male and female Bmal1 knockout mice. b) Distribution of the gene fraction of posterior amplitude samples being larger in female Bmal1 knockout cell types relative to male. c) Comparison of $\log_{10}$ circadian Bayes factors between male and female major cell types after Bmal1 knockout.

Supplementary Figure 11: Sex and Bmal1 knockout effects on individual gene expression waveforms in SMCs. a) Posterior circadian waveforms in SMCs for representative genes with sex-dependent Bmal1 knockout effects. b) Assessing similarity between confident cycling gene expression in males and females, in the Bmal1 WT and KO SMCs. c) The fraction of confident cycling genes with more similar likelihood between the sexes in the KO conditions over the WT conditions, at various $\log_{10}$ Bayes factor thresholds.

Supplementary Figure 12: Sex and Bmal1 knockout effects on individual gene expression waveforms in SMCs of the ascending aorta/aortic arch and macrophages. a) Posterior circadian waveforms in SMCs of the ascending aorta/aortic arch for representative genes with sex-dependent Bmal1 knockout effects. b) Posterior circadian waveforms in macrophages for representative genes with sex-dependent Bmal1 knockout effects.

Supplementary Figure 13: General characteristics of the sex and misalignment effect on cycling expression. a) Number of confident circadian cycling genes in major cell types of aorta from male and female misaligned mice. b) Distribution of the gene fraction of posterior amplitude samples being larger in female misaligned cell types relative to male misaligned cell types, considering the union of confident cycling genes.

Supplementary Figure 14: Comparison of female aligned vs. female misaligned cell type rhythmicity. Within each cell type and time point, the number of cells and pseudobulk UMIs were downsampled to match between female aligned and misaligned mice. The number of called cyclers across $\log_{10}$ circadian Bayes factor thresholds for major cell types and SMC subtypes.

Supplementary Figure 15: Male vs. female entrainment speed of major aortic cell types after acute misalignment. For each cell type, posterior distributions of the acrophase delay after misalignment were computed for the male and female genes. The distribution of the fraction of posterior samples that were smaller in females was visualized for a) SMCs and fibroblasts, and b) SMC subtypes.

Supplementary Figure 16: Acute misalignment and sex effects on SMC phenotypic switching genes. a) Posterior circadian waveforms of *Id1* and *Id3* in male and female SMCs, comparing aligned and misaligned conditions. b) Posterior circadian waveforms of *Ccn2* in male and female SMCs, comparing aligned and misaligned conditions.

Supplementary Figure 17: Acute misalignment and sex effects on endothelial and macrophage gene expression. a) Posterior circadian waveforms of key EC genes (*Egr3*, *Nrarp*, *Fabp4*) in male and female ECs, comparing aligned and misaligned conditions. b) Posterior circadian waveforms of key macrophage genes (*Hbegf*, *Dusp2*, *S100a6*) in male and female macrophages, comparing aligned and misaligned conditions.